

September 2001
Final



Western Forest Products Limited

Tree Farm Licence 6

Quatsino Sound – North Vancouver Island

Sustainable Forest Management Plan

MANAGEMENT PLAN 9

November 1, 2001 to October 31, 2006

*This Plan also serves as the Management Plan and Commitment for
Managed Forest 29 and 31*

Approved – July 23, 2001

The Environmental Management System covering forest operations in TFL 6 is registered as conforming to ISO 14001 standards. This certification was achieved in April 2000 through the Canadian Quality Management Institute.





Western Forest Products Limited

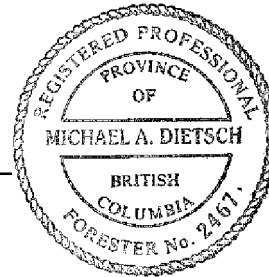
Tree Farm Licence 6

Management Plan 9

This Management Plan was prepared
by and under the supervision of

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Mike Dietsch, R.P.F.
Regional Forester



And submitted on behalf of
Western Forest Products Limited by

A handwritten signature in black ink, appearing to read 'W. E. Dumont'.

W.E. Dumont, R.P.F.
Chief Forester



TFL 6 Management Plan Preparation Component and Timeline Summary

Component	Date
• Licensee advertises for public comment • Licensee makes current MP available for review	May 30/1998
• Licensee submits review strategy (if not in current MP)	July 31/1998
• RM accepts or rejects review strategy. • RM sends pre-SMOOP assessment & guidelines • Licensee submits comments from current MP review.	Sept 30/1998
• Licensee prepares the SMOOP & objectives for employment & economic opportunity (EEO) & refers them as per review strategy.	Sept 30/1998 To June 30/1999
• Licensee submits SMOOP & objectives for EEO, with comments from referral, to the RM.	July 30/1999
• RM accepts or rejects SMOOP	Aug 31/1999
• TS Forester sends notice with information requirements	Sept 30/1999
• Licensee submits Timber Supply Analysis Information Package to the Timber Supply Forester	Nov 31/1999
• Timber Supply Forester accepts or rejects the Timber Supply Analysis Information Package	June 13/2000
• Licensee submits a Timber Supply Analysis (TSA) to the Timber Supply Forester. • Licensee submits a 20-year plan to the DM.	Sept 8/2000
• Licensee submits draft MP & draft proposals for EEO to the RM. • Licensee refers draft MP & draft proposals for EEO as per review strategy.	Oct 31/2000
• Timber Supply Forester accepts or rejects TSA. • District Manager accepts or rejects 20-year plan.	Nov 30/2000
• RM provides comments on draft MP to Licensee.	Dec 31/2000
• Licensee submits proposed MP & proposals for EEO to the Chief Forester, with input received from the referrals.	May 1/2001
• Chief Forester approves or rejects MP.	Oct 31/2001

Revised April 2001

Table of Contents

1.0	INTRODUCTION	1
1.1	PURPOSE.....	1
1.2	DESCRIPTION OF THE TFL.....	1
1.3	HISTORY AND PROGRESS.....	4
1.4	LICENCE HOLDER AND ADMINISTRATION	8
2.0	SUSTAINABLE FOREST MANAGEMENT PRINCIPLES	9
2.1	SUSTAINABLE FOREST MANAGEMENT POLICY	9
2.2	LINKING WFP'S SUSTAINABLE FOREST MANAGEMENT PRINCIPLES TO CRITERIA AND INDICATORS	11
2.2.1	<i>Hierarchy of SFM Levels</i>	<i>11</i>
2.2.2	<i>Sustainable Forest Management SFM system</i>	<i>12</i>
2.3	WFP'S FOREST CERTIFICATION INITIATIVES	13
3.0	FOREST MANAGEMENT INVENTORY AND PLANNING	14
3.1	RESOURCE INVENTORIES	14
3.1.1	<i>Forest Cover</i>	<i>15</i>
3.1.2	<i>Environmentally Sensitive Areas</i>	<i>15</i>
3.1.3	<i>Operability.....</i>	<i>15</i>
3.1.4	<i>Recreation.....</i>	<i>15</i>
3.1.5	<i>Visual Landscape.....</i>	<i>17</i>
3.1.6	<i>Terrain Stability</i>	<i>17</i>
3.1.7	<i>Ownership and other Tenures</i>	<i>18</i>
3.1.8	<i>Archaeological Resource Potential.....</i>	<i>18</i>
3.1.9	<i>Ecosystem Mapping.....</i>	<i>18</i>
3.1.10	<i>Streams.....</i>	<i>19</i>
3.1.11	<i>Wildlife</i>	<i>19</i>
3.1.12	<i>High Conservation Value Forests.....</i>	<i>20</i>
3.2	PLANNING LINKAGES - STRATEGIC AND OPERATIONAL PLANNING	21
3.2.1	<i>Regional Plans</i>	<i>22</i>
3.2.2	<i>Landscape Unit Plans</i>	<i>23</i>
3.2.3	<i>Forest Development Plans</i>	<i>23</i>
3.2.4	<i>Stand Level Prescriptions.....</i>	<i>24</i>
3.2.5	<i>Adaptive Management</i>	<i>24</i>
3.2.6	<i>Ecosystem Based Management.....</i>	<i>25</i>
4.0	SUSTAINABLE FOREST MANAGEMENT STRATEGY	26
4.1	ECONOMICALLY VIABLE FOREST MANAGEMENT.....	26
4.1.1	<i>Multiple Benefits.....</i>	<i>26</i>
4.1.1.1	<i>Timber Supply</i>	<i>26</i>
4.1.1.2	<i>Harvest Methods</i>	<i>28</i>
4.1.1.3	<i>Employment</i>	<i>29</i>
4.1.1.4	<i>Contractor Commitment.....</i>	<i>30</i>
4.1.1.5	<i>Profitability.....</i>	<i>31</i>
4.1.1.6	<i>Non-Timber Forest Products.....</i>	<i>31</i>
4.1.1.7	<i>Minor Forest Products.....</i>	<i>33</i>
4.1.1.8	<i>Access Management.....</i>	<i>34</i>
4.1.1.9	<i>Recreation</i>	<i>34</i>
4.1.1.10	<i>Research and Development</i>	<i>36</i>
4.1.1.11	<i>Contribution to Provincial Revenues</i>	<i>37</i>

4.2	ENVIRONMENTALLY APPROPRIATE FOREST MANAGEMENT	37
4.2.1	<i>Conservation of Biological Diversity</i>	37
4.2.1.1	Ecological Diversity	38
4.2.1.2	Species Diversity	41
4.2.1.3	Genetic Diversity	43
4.2.2	<i>Maintenance of Ecosystem Condition and Productivity</i>	44
4.2.2.1	Forest Health	44
4.2.2.2	Forest Ecosystem Resilience	47
4.2.2.3	Forest Ecosystem Productivity	49
4.2.3	<i>Conservation of Soil and Water Resources</i>	51
4.2.3.1	Productive Area	51
4.2.3.2	Water Quality	52
4.2.4	<i>Contribution to Global Ecological Cycles</i>	53
4.2.4.1	Carbon Sequestration	53
4.2.5	<i>High Conservation Value Forests</i>	54
4.3	SOCIALLY BENEFICIAL FOREST MANAGEMENT	54
4.3.1	<i>Responsibility for Sustainable Development</i>	54
4.3.1.1	First Nations	54
4.3.1.2	Communities	56
4.3.1.3	Employment	58
4.3.1.4	Compliance	59
4.3.1.5	Employee Relations	59
4.3.1.6	Safety	60
5.0	SUSTAINABLE FOREST MANAGEMENT PLAN CONSULTATION	61
6.0	SUMMARY IMPACT ANALYSIS	62
6.1	ECONOMIC	62
6.2	ENVIRONMENTAL	63
6.3	SOCIAL	64
6.4	IMPACTS DUE TO IMPLEMENTATION OF MANAGEMENT PLAN 9	65
6.4.1	<i>Harvest and Employment Levels</i>	65
6.4.2	<i>Non-Timber Values</i>	65
6.4.3	<i>General Impacts of Implementation</i>	65

Tables

TABLE 1	TFL LANDBASE STATUS.....	2
TABLE 2	RESOURCE INVENTORY STATUS	14
TABLE 3	RECREATIONAL SITES AND TRAILS IN AND AROUND TFL 6: 1995 – 1999	16
TABLE 4	TFL 6 ECOSYSTEM CLASSIFICATION SYSTEM	18
TABLE 5	RESOURCE MANAGEMENT ZONES AND REGIMES	22
TABLE 6	LUS AND BEOs FOR TFL LANDBASE	23
TABLE 7	CUT CONTROL PERFORMANCE 1950-1999	26
TABLE 8	DISTRIBUTION OF LOGGING SYSTEMS	29
TABLE 9	DIRECT EMPLOYMENT 1995 - 2000	29
TABLE 10	COMPLIANCE WITH TIMBER HARVESTING CONTRACTING REGULATION	30
TABLE 11	RECREATION ACTIVITIES SUMMARY – ESTIMATED USER DAYS, TFL 6 1995-1999	36
TABLE 12	RED AND BLUE LISTED ECOSYSTEMS POTENTIALLY IN NVIR	39
TABLE 13	RED, BLUE LISTED AND REGIONALLY IMPORTANT MAMMALS AND BIRDS	42
TABLE 14	COMMON INSECTS AND DISEASE FOUND IN TFL 6	45
TABLE 15	FIVE-YEAR BASIC SILVICULTURE ACTIVITY GOALS (HA) FOR TFL 6.....	49
TABLE 16	FIVE-YEAR ENHANCED SILVICULTURE GOALS (HA).....	50
TABLE 17	ANNUAL DIRECT EMPLOYMENT 1995 – 2000	58
TABLE 18	INDICATORS OF ECONOMICALLY VIABLE FOREST MANAGEMENT	62
TABLE 19	INDICATORS OF ENVIRONMENTALLY APPROPRIATE FOREST MANAGEMENT	63
TABLE 20	INDICATORS OF SOCIALLY BENEFICIAL FOREST MANAGEMENT	64

Figures

FIGURE 1	TFL 6	1
FIGURE 2	STATUS MAP	3
FIGURE 3	COMMUNITIES	4
FIGURE 4	FOREST MANAGEMENT CYCLE	5
FIGURE 5	TIMBER UTILIZATION	8
FIGURE 6	MAJOR RECREATION SITES AND TRAILS.....	17
FIGURE 7	UNGULATE WINTER RANGES	19
FIGURE 8	RESOURCE MANAGEMENT ZONES OF TFL 6	22
FIGURE 9	LANDSCAPE UNITS OF TFL 6.....	23
FIGURE 10	TIMBER HARVESTING PROCESS.....	28
FIGURE 11	TRAPLINES.....	32
FIGURE 12	KNOWN AQUATIC LICENCES	33
FIGURE 13	PROVINCIAL PARKS, RESERVES AND RECREATIONAL SITES	35
FIGURE 14	TREE IMPROVEMENT AND REFORESTATION CYCLE	44
FIGURE 15	DRINKING WATER SOURCES	52
FIGURE 16	FIRST NATIONS TERRITORIES.....	55
FIGURE 17	FOREST OPERATIONS	58

Appendices

APPENDIX I	TFL AMENDMENTS
APPENDIX II	TIMBER SUPPLY INFORMATION PACKAGE
APPENDIX III	TIMBER SUPPLY ANALYSIS
APPENDIX IV	SILVICULTURE PROJECT HISTORY
APPENDIX V	SILVICULTURE PRESCRIPTION SAMPLE
APPENDIX VI	MANAGED FOREST NO. 29 AND 31
APPENDIX VII	SAMPLE FIRE PREPAREDNESS PLAN
APPENDIX VIII	STANDARD OPERATING PROCEDURES – ROAD DEACTIVATION
APPENDIX IX	FOREST RESEARCH PROJECTS
APPENDIX X	PUBLIC REVIEW PLAN – MP 9
APPENDIX XI	TRAILS AND RECREATION SITES
APPENDIX XII	NORTH VANCOUVER ISLAND REGION VISITORS GUIDE
APPENDIX XIII	TFL 6 SILVICULTURE STANDARDS
APPENDIX XIV	HABITAT RESTORATION PLANS & KEOGH WATERSHED RESTORATION BROCHURE
APPENDIX XV	TWENTY-YEAR PLAN
APPENDIX XVI	ISO 14001 REGISTERED ENVIRONMENTAL MANAGEMENT SYSTEM
APPENDIX XVII	PROFILES OF MAIN COMMUNITIES
APPENDIX XVIII	DOMAN INDUSTRIES ANNUAL STATUTORY REPORT
APPENDIX XIX	WESTERN SPIRIT AND ENVIRONMENT MATTERS
APPENDIX XX	HIGH CONSERVATION VALUE FOREST – NORTH VANCOUVER ISLAND
APPENDIX XXI	HIGH CONSERVATION VALUE FOREST ATTRIBUTE MANAGEMENT STRATEGY
APPENDIX XXII	STRATEGY OUTLINE FOR MANAGING WILDLIFE
APPENDIX XXIII	STATEMENT OF MANAGEMENT OBJECTIVES, OPTIONS AND PROCEDURES
APPENDIX XXIV	STATEMENT OF OBJECTIVES FOR EMPLOYMENT AND ECONOMIC OPPORTUNITY

**A CD folio of all maps found in this Plan is enclosed
for larger scale review**

1.0 INTRODUCTION

1.1 Purpose

The Tree Farm Licence (TFL) agreement requires the Licensee to submit for Ministry of Forests (MoF) approval a Management Plan. Revised every five years, the Management Plan provides objectives, goals and strategies to guide management activities until the next revision. The Management Plan and associated Timber Supply Analysis are used by the MoF Chief Forester to determine the Allowable Annual Cut (AAC). The Management Plan provides direction for the preparation of operational plans. The Plan is focused on results and outcomes so that the Licensee, Western Forest Products Limited (WFP) can evaluate performance and continuously improve management practices. The Plan contains measurable parameters in the form of indicators that will monitor the achievement of sustainable forest management objectives for the period November 1, 2001 to October 31, 2006. It is also intended to form the foundation for Forest Stewardship Council and Canadian Standards Association sustainable forestry certifications. It is expected that further amendments may be proposed for this plan to fully address FSC and CSA requirements for forest certification

1.2 Description of the TFL

The Quatsino Tree Farm Licence 6 (TFL 6) is located on northern Vancouver Island in the vicinity of Quatsino Sound (Figure 1). The total area of the TFL is approximately 198,113 hectares of which 156,769 hectares is Crown land and 41,334 hectares are timber licences or fee simple lands owned by WFP. A detailed summary of the TFL land status and mature timber volumes is in Table 1. The timber harvesting landbase is 149,747 ha. Figure 2 shows the ownership associated with TFL 6. The fee simple (private) lands and lands controlled by the licensee are encompassed in Managed Forests (MF) 29 and 31. Maps and area summaries for MF 29 and 31 are found in Appendix VI. The TFL is made up of two geographically separated management areas. Block 1 is the largest management unit and includes watersheds that drain westward into Quatsino Sound or the Pacific Ocean between Quatsino Sound and San Josef Bay. Block 2 includes watersheds that drain northeastward into Queen Charlotte Strait between Fort Rupert and Port McNeill plus those that drain westward into the Waukwaas watershed and Rupert Arm.



Figure 1 – TFL 6



Table 1 – TFL Landbase Status
TIMBER HARVESTING LAND BASE DETERMINATION FOR TFL 6

Classification	Area (ha)		Total	Mature Volume (m ³)		
	Schedule A	Schedule B		Schedule A	Schedule B	Total
Total Area	41,344.4	156,768.9	198,113.3	15,623,738.4	48,786,233.6	64,409,972.0
Less: Non-Forest	1,864.1	10,291.0	12,155.1	0	0	
Less: Non-Productive Forest	72.5	394.8	467.3	0	0	
Total Productive Forest	39,407.8	146,083.1	185,490.9	15,623,738.4	48,786,233.6	64,409,972.0
Less Reductions to Total Productive Forest:						
Non-Commercial (NP Br)	78.8	335.9	414.7	0	0	0
Low Site	1,851.3	10,096.1	11,947.4	949,560.0	4,729,139.2	5,678,699.2
Inoperable / Inaccessible (I, Oce, Ohe)	1,771.7	10,807.6	12,579.3	1,508,351.6	9,113,339.1	10,621,690.6
Riparian Reserves	1,576.6	4,725.7	6,302.3	592,829.4	1,058,842.6	1,651,672.0
Wildlife Habitat Reserves (e.g. UWR)	402.9	1,273.9	1,676.8	360,263.7	986,753.4	1,347,017.1
Recreational Areas	22.8	157.3	180.1	17,178.2	138,279.0	155,457.2
Ecological Reserve	0.0	120.0	120.0	0	52,578.0	52,578.0
Problem Forest Types (Deciduous)	0.0	0.0	0.0 ¹	348.4	731.1	1,079.6
Unclassified Roads, Trails and Landings	556.8	1,966.7	2,523.5	64,773.7	114,190.0	178,963.7
Total Reductions to Productive Forest	6,260.7	29,483.2	35,743.9	3,493,305.0	16,193,852.4	19,687,211.4
Less: Volume Reductions						
Total Reduced Land Base	33,147.1	116,599.9	149,747.0	12,130,433.4	32,592,381.2	44,722,814.6
Less: Not Sufficiently Restocked Areas	684.3	1,904.0	2,588.3	0	0	0
Add: Not Sufficiently Restocked Areas	684.3	1,904.0	2,588.3	0	0	0
Timber Harvesting Land Base	33,147.1	116,599.9	149,747.0	12,130,433.4	32,592,381.2	44,722,814.6
Less: Future Roads, Trails and Landings	756.4	2081.3	2,837.7	0	0	0
Total Long Term Land Base	32,390.7	114,518.6	146,909.3	12,130,433.4	32,592,381.2	44,722,814.6

¹ Deciduous leading stands contribute to the timber harvesting landbase, but only their conifer volume is applied to timber supply

The topography within the TFL is variable. Relatively low relief, undulating terrain characterizes the easterly and westerly sections of the TFL. The central and inland portions of the TFL are characterized as mountainous and steep. Numerous rivers and streams drain the licence area. Most streams support significant anadromous and resident fish populations. Large animals, notably, Columbia black-tailed deer, cougars and black bears are abundant throughout the licence area. Numerous other large and small mammals, amphibians, fish and birds can also be found. These are identified in Appendix XXII.

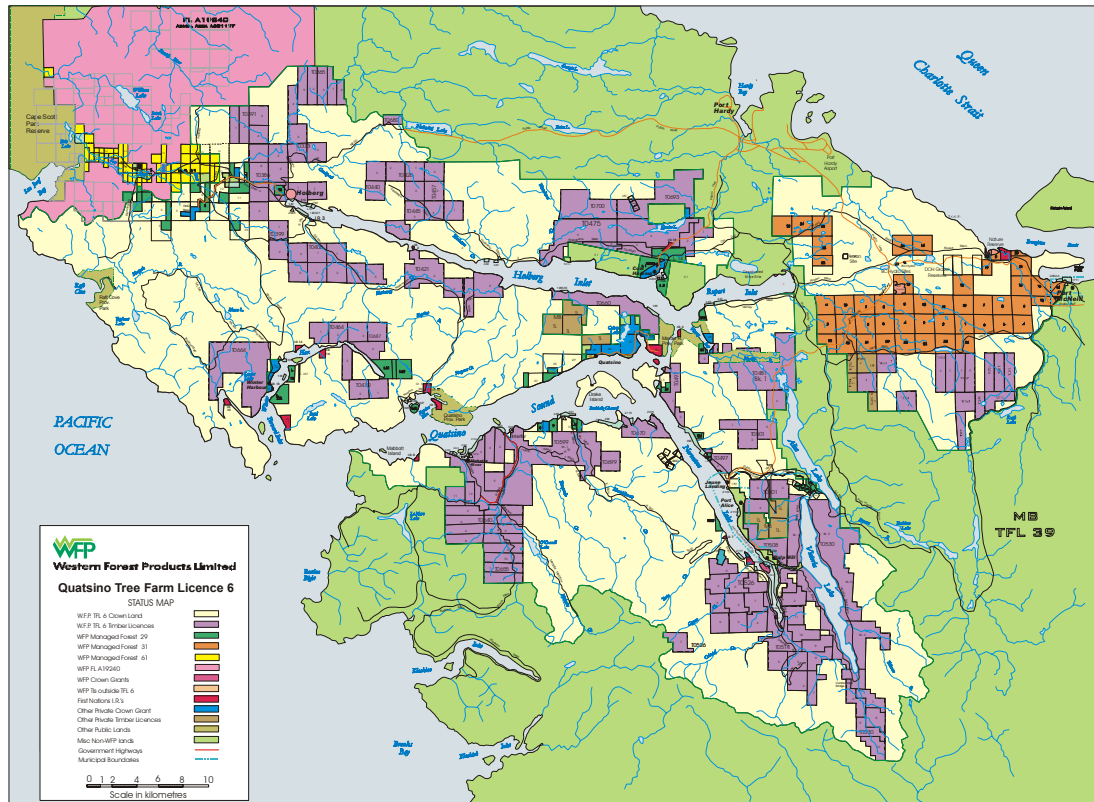


Figure 2 – Status Map

The forests of TFL 6 lie within the wetter maritime Coastal Western Hemlock biogeoclimatic zone. Annual precipitation levels reach 3,000 to 5,000 mm. The climate is characterized by mild, wet winters with daily mean minimum temperatures December to February of 0 to 2 degrees C. Summers are generally cool and moist with mean daily maximum temperatures of 18 to 20 degrees C during July and August. However, local climates within the TFL can be significantly different due to topographic influences and the movement of low cloud and fog from offshore onto northern Vancouver Island. The dominant timber species is western hemlock, which occurs in conifer stands mixed with varying amounts of amabilis fir and western red cedar. Lesser amounts of Sitka spruce, yellow cedar, Douglas fir, alder, shore pine and mountain hemlock also occur.

There are seven communities within or adjacent to the licence area. Their livelihood and stability depends mostly or in part on the economic activities generated on the North Island. These are the communities of Port McNeill, Port Alice, Port Hardy, Holberg, Coal Harbour, Winter Harbour, and Quatsino as noted in Figure 3. WFP harvesting operations in TFL 6 and adjacent forest licences, and the associated dissolving pulp manufacturing facility in Port Alice, are the major employment source for north Vancouver Island. There is also economic activity from other forest licensees, fish farming, commercial and recreational fishing and expanding tourism.

Port McNeill and Port Hardy are connected to southern Vancouver Island by Highway 19. Port Alice and Coal Harbour are served by paved roads connected to Highway 19. Holberg and Winter Harbour are served by a gravel road connected to Highway 19 south of Port Hardy. Quatsino is accessible by water from Coal Harbour. This community currently does not have nor does it desire road access.

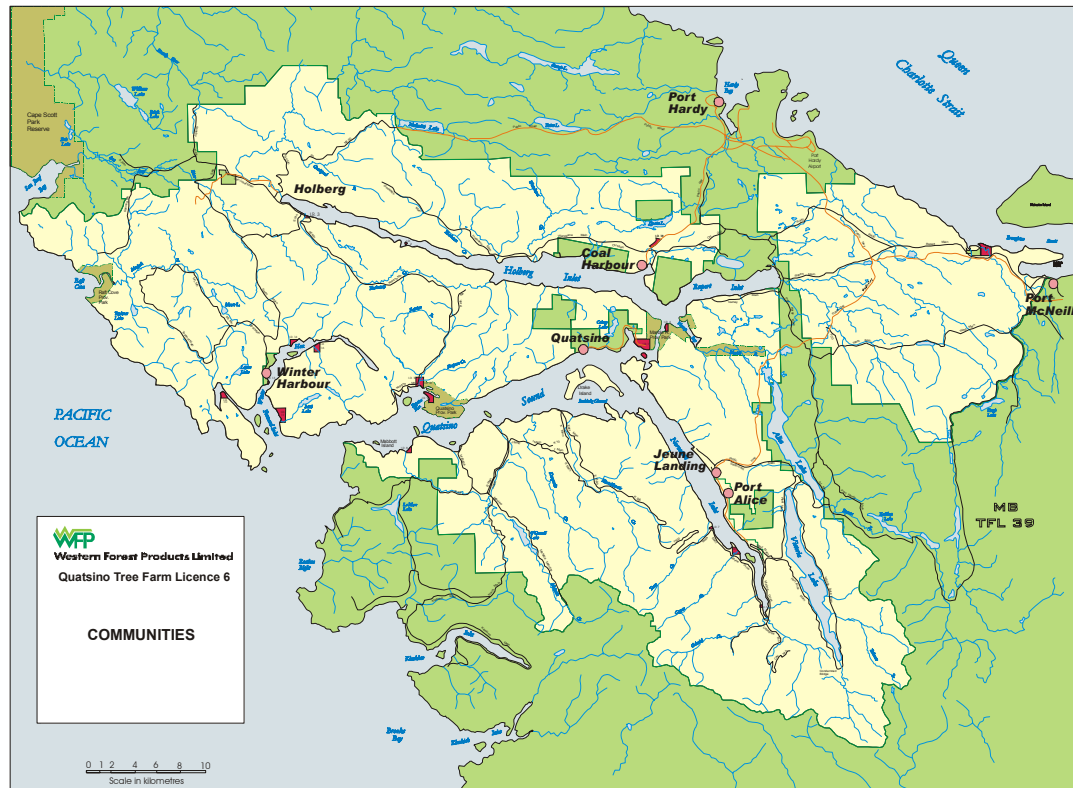


Figure 3 – Communities

1.3 History and Progress

TFL 6 is held and managed by Western Forest Products Limited (WFP). Western Forest Products Limited is a wholly owned subsidiary of Doman Industries Limited.

Block 1 is the original TFL 6 area granted to Western's predecessor, British Columbia Pulp & Paper Company Limited as Quatsino Forest Management Licence 6 on October 26, 1950. Block 2 is the former Block 4 of Tree Farm Licence 25 which was granted to another predecessor company, Alaska Pine and Cellulose Limited, on May 21, 1958. On October 1, 1998 these two areas were consolidated into a new TFL 6 made up of Block 1 (former TFL 6) and Block 2 (former TFL 25 Block 4). Now in its fifth decade of operation, WFP and predecessors have responsibly managed the licence for its long-term sustainability. WFP is proud of the efforts of our employees and contractors in helping to sustain this important generator of employment and crown revenues. Appendix I lists amendments to the licence area that have occurred since 1950.

The following accomplishments indicate the licensee's willingness and ability to manage more than 198,000 hectares of lands and forests in Tree Farm Licence 6 in a spirit of co-operation with government agencies, its employees and a multi-faceted public. Although site specific management cycles will vary considerably, Figure 4 illustrates a generalized Forest Management Cycle.

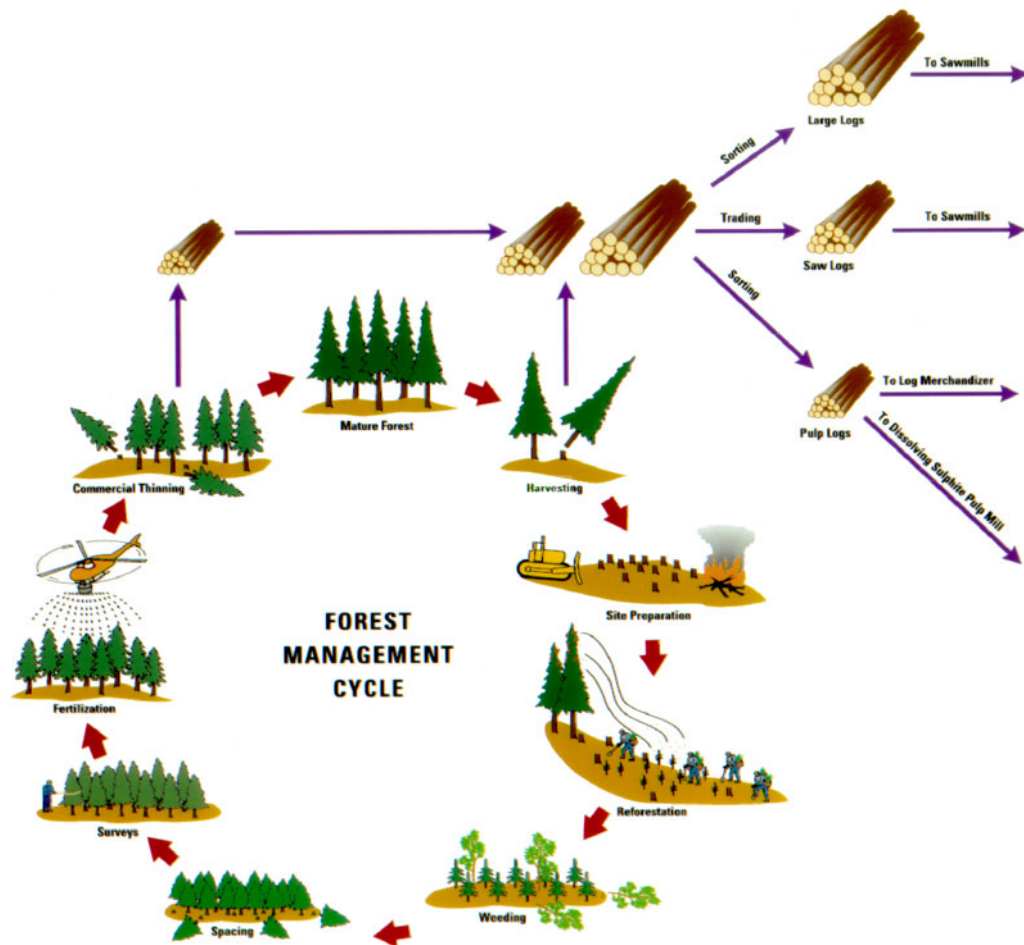


Figure 4 – Forest Management Cycle

Biodiversity Conservation

- A detailed summary of biodiversity has been completed for TFL 6.
- WFP has worked with government over the last three years in establishing Landscape Unit boundaries and draft biodiversity emphasis options within the TFL.
- Analysis of current seral stage distribution within the TFL has been completed to determine the amount and biological significance of old growth remaining in the TFL.
- WFP is committed to working with government and stakeholders to complete biodiversity planning as part of the Landscape Unit Planning process over the next two years.
- WFP has completed an initial assessment of the TFL and adjacent tenures for the presence of High Conservation Value Forest attributes.
- Management strategies are being developed to maintain and where possible enhance High Conservation Value Forest attributes (including biodiversity) within the TFL.

First Nations

- First Nations review of planning, harvesting and silviculture activities is in place.
- The development of First Nations silviculture work capacity has been supported by coordinating training and the direct award of silviculture contracts for spacing, pruning, fire suppression and mop-up, brushing and creek cleaning to members of the Quatsino First Nation and the Kwakiutl First Nation. Coordinating training and providing employment for Tlatlasikwala First Nation members within the company planting program.
- The licensee remains committed to continue developing First Nations capacity in forest

management.

- The licensee supports a First Nations Coordinator and review of forest harvesting is undertaken in conjunction with Forest Development Plans.
- A protocol is being developed with the three Bands whose territory covers the TFL in respect of silviculture contracting and FRBC projects.

Communities and Employment

- Forestry and manufacturing jobs related to the TFL 6 timber harvest are critically important to the communities of Port McNeill, Port Alice, Holberg, Winter Harbour, Coal Harbour and Port Hardy. Secondary jobs are also maintained in Campbell River and Courtenay as well other Vancouver Island and southern coastal communities. In 1998, 968 direct person years of employment were created by forest management in TFL 6. Poor markets and high inventories in 1998 resulted in the licensee achieving only 69% of the AAC. Doman Industries' sawmills also experienced significant down time in 1998 and 1999. Consequently, the current person days of employment are lower than historical averages. In 1999, 1273 direct person years of employment were maintained when 96% of the AAC was achieved.
- The Licensee has been a long time supporter of North Island communities. Open Houses are held each year to present logging and silvicultural plans, company staff are active on community boards and in local government and the company has been a vital part of community development including providing donations for local causes, sponsoring sports teams, sponsoring community events, and developing infrastructure (arena, pool, curling rink, golf course, track, etc.).

Recreation

- WFP maintains 15 trails and recreation sites on both private and public land within and adjacent to the TFL. Many of the sites are managed cooperatively with the Ministry of Forests. In 1999 there were more than 64,000 person-days of recreational use at these sites.
- Staff participated in a FRBC study prepared by the Economic Planning Group for the Regional District of Mount Waddington titled, "Northern Vancouver Island Forest Recreation and Tourism Opportunities Study".
- Each year hundreds of TFL visitors participate in guided tours to view harvesting and forestry activities.
- The recreation inventory for TFL 6 was reviewed in 1999. As there were no significant changes in the recreational use of TFL lands, an update to the inventory was not necessary for MP 9.

Timber Harvesting

- Over the last 49 years the licensee has harvested over 99% of the available Allowable Annual Cut and has been at all times in compliance with the cut control regulations.
- Harvesting across the timber profile has been maintained.
- Non-conventional harvesting techniques such as long-line and helicopter logging have been utilized.
- Compliance with contracting regulations has always been met.

Small Business Forest Enterprise Program

- In 1988, the Forest Act was amended and TFL volume was removed from TFL licensee control for a small business logging program. Further amendments were made as a consequence of Doman Industries assuming full control of WFP in 1988. In 1995 an agreement was reached between the Ministry of Forests and Western Forest Products

Limited, for WFP to relinquish a Forest Licence in the nearby Strathcona Timber Supply Area and delete a small area from TFL 6. In return WFP received the volume in TFL 6 that had been allocated to the Small Business Forest Enterprise Program. This agreement simplified administration of TFL 6 and increased the employment security of WFP employees and contractors. There still remains some unharvested volume from this program and this will be monitored by WFP during the term of MP 9.

- The program is allocated an annual volume of 13,242 m³ from the AAC of TFL 6.
- The Port McNeill Forest District manager is responsible for administering the program.
- Operating areas for the small business program are identified and are located in several drainages.
- The majority of small business wood is sold through value added bid proposal timber sale licences.

Basic Silviculture

- Reforestation has kept pace with harvesting since the TFL was granted in 1950.
- The licensee has logged about 73,000 ha. Planting has been undertaken on 41,000 ha. More than 38 million seedlings of 11 different species have been planted. Natural regeneration occurs readily and has accounted for in excess of 32,000 ha of re-stocking.
- Tree improvement programs for Douglas Fir, western hemlock and Sitka spruce began in 1959, 1966 and 1969 respectively. Today all of the Douglas fir and 80% of the western hemlock and most of the red cedar seedlings planted are grown from seed orchard seeds. No genetically modified seed or trees have been used for reforestation.
- The Company maintains a 3 million-tree seedling forest nursery in Saanich.
- There is currently no backlog of not satisfactorily restocked (NSR) area on the TFL.
- Appendix IV provides a history of silviculture projects completed.

Enhanced Silviculture

- More than 10,832 ha of second growth stands have been juvenile spaced to accelerate tree size development by concentrating growth onto less trees.
- Fertilization to enhance forest productivity has occurred on more than 9,715 ha.
- 2,766 ha have been pruned to improve future wood quality.

Research

- The Company maintains several tree improvement and reforestation research sites (Appendix IX) throughout TFL 6. These are maintained in cooperation with the BC Forest Service (BCFS), the Canadian Forestry Service (CFS) and/or other Licensees.
- WFP is a founding member and continues to be a strong supporter of the Salal Cedar Hemlock Integrated Research Program.
- WFP also provides sites to the Forest Engineering Research Institute of Canada (FERIC) and BCFS researchers for various tree improvement, silviculture, harvesting and road building trials.
- The licensee's predecessor established the Saanich Forestry Center in 1964 to provide improved planting stock for reforestation. In 1998 WFP's original tree improvement program was transferred from its Lost Lake location to the Saanich Forestry Centre. Seed orchards for western hemlock, western red cedar and Sitka spruce are maintained. Yellow cedar hedges are also maintained for steckling production. WFP is a member of the Northwest Hemlock Tree Improvement Cooperative Program (HEMTIC).
- WFP has established several minor trials within the licence area. Some of the trials include time of planting fertilization, variable spacing trials, juvenile spacing methods, herbicide treatment, brush blankets, and drainage.

1.4 Licence Holder and Administration

Western Forest Products Limited is a wholly owned subsidiary of Doman Industries Limited (Doman). Doman is a coastal B.C. forest products company involved in timber harvesting, saw milling, value-added lumber remanufacturing, dissolving sulphite pulp production and kraft pulp production. Doman's nine saw mills have an annual lumber production capacity of 1.1 billion board feet. Major markets include the U.S.A., Japan and Europe. Doman also owns one sulphite pulp mill at Port Alice, which has an annual production capacity of 160,000 air-dried metric tonnes (ADMT). Doman has a kraft pulp mill at Woodfibre with an annual production capacity of 240,000 ADMT. Major markets for pulp include North America, Europe and the Far East.

Western Forest Products Limited is responsible for timber harvesting and reforestation on approximately 885,000 ha of Crown and private forest lands located in coastal British Columbia. WFP has an annual timber harvest of approximately 4.2 million cubic metres under its tenures, which include three Tree Farm Licences, seven Forest Licences, 5 Managed Forests and 125 Timber Licences. The Company's AAC together with log trading and other log supply agreements provides for approximately 85% of the fibre requirements for the Company's manufacturing plants. The remainder of the wood is purchased on the open market.

WFP's forest operations are organized into three regions. North Vancouver Island Region, with a regional office in Port McNeill, administers harvesting and reforestation activities in the north end of Vancouver Island. Mainland/Islands Region, with a regional office in Campbell River, administers harvesting and reforestation activities in numerous north and central coast operations as well as operations on the Queen Charlotte Islands, southern Vancouver Island and the lower mainland coast. Nootka Region, with a regional office in Gold River, administers harvesting and reforestation activities on the central, west coast of Vancouver Island. WFP's corporate head office is located in Vancouver. Appendix XVIII contains a copy of the Corporate Annual Statutory Report for 2000. The company's overall utilization of forest products is illustrated in Figure 5.

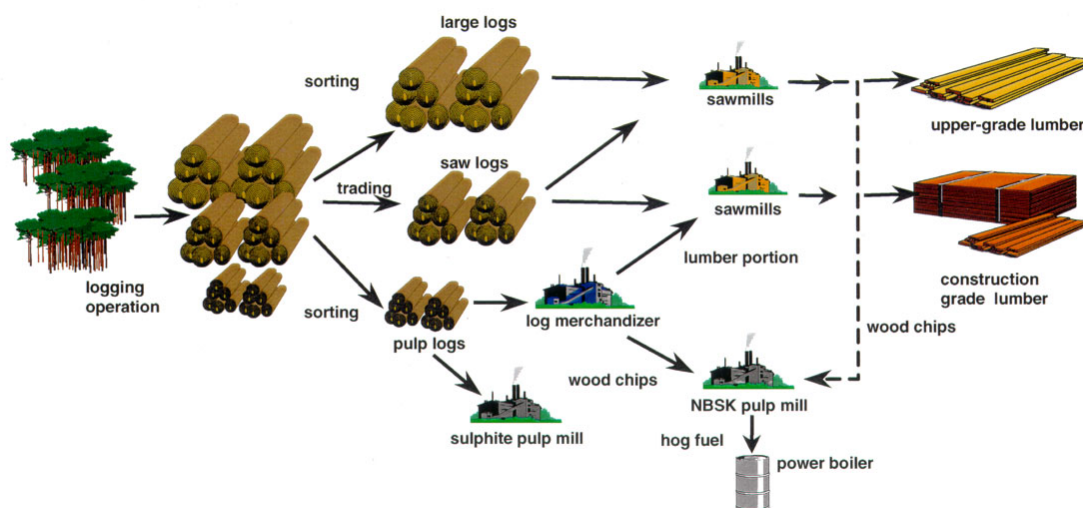


Figure 5 – Timber Utilization

2.0 SUSTAINABLE FOREST MANAGEMENT PRINCIPLES

2.1 Sustainable Forest Management Policy

WFP's forest management is guided by three overarching principles. These are:

- Our operations must be economically viable. Our company must operate in a financially successful manner to meet shareholders expectations of a fair return on investment and to meet social and environmental commitments;
- Our actions must be socially beneficial to local, regional and First Nation's communities. WFP is committed to respecting, understanding and supporting First Nations, community and employee aspirations for stability and certainty;
- Our activities must be environmentally appropriate. WFP is committed to the protection of the environment and the sustainable development of the resources under our stewardship through sound forestry and environmental management practices that meet or exceed government standards.
- Under these principles, our company is committed to:
 - Establishing harvest levels that balance the carrying capacity of forest ecosystems with social, environmental, and economic considerations.
 - Utilizing forest practices appropriate to maintaining, social, environmental, and economic objectives
 - Engaging a team of dedicated professional foresters and planners committed to implementing and practicing sustainable forest management
 - Continually enhancing our ability to plan and manage by promoting applied research and using leading edge inventory and modeling systems.
 - Planning for the long-term in an integrated manner to incorporate the full range of forest values including soil, water, fish and wildlife, archaeological, scenic resources and biological diversity.
 - Maintaining a practical, registered management system to guide and direct environmental aspects of company operations
 - Performing regular, internal and external audits to ensure compliance with forest practices legislation and the commitments made in this SFM statement.
 - Committing to a level of compliance that meets or exceeds legal requirements and satisfies public expectations
 - Continuously improving all aspects of our forest practices through adaptive management.
 - Communicating our performance to stakeholders and the public on a regular basis
 - Maintaining biological diversity and conservation objectives at regional, landscape and stand levels.
 - Sustaining, enhancing and protecting forest ecosystem functions.
 - Maintaining soil productivity and water quality by harvesting with environmentally appropriate systems.
 - Researching habitat needs of fish and wildlife species and populations and

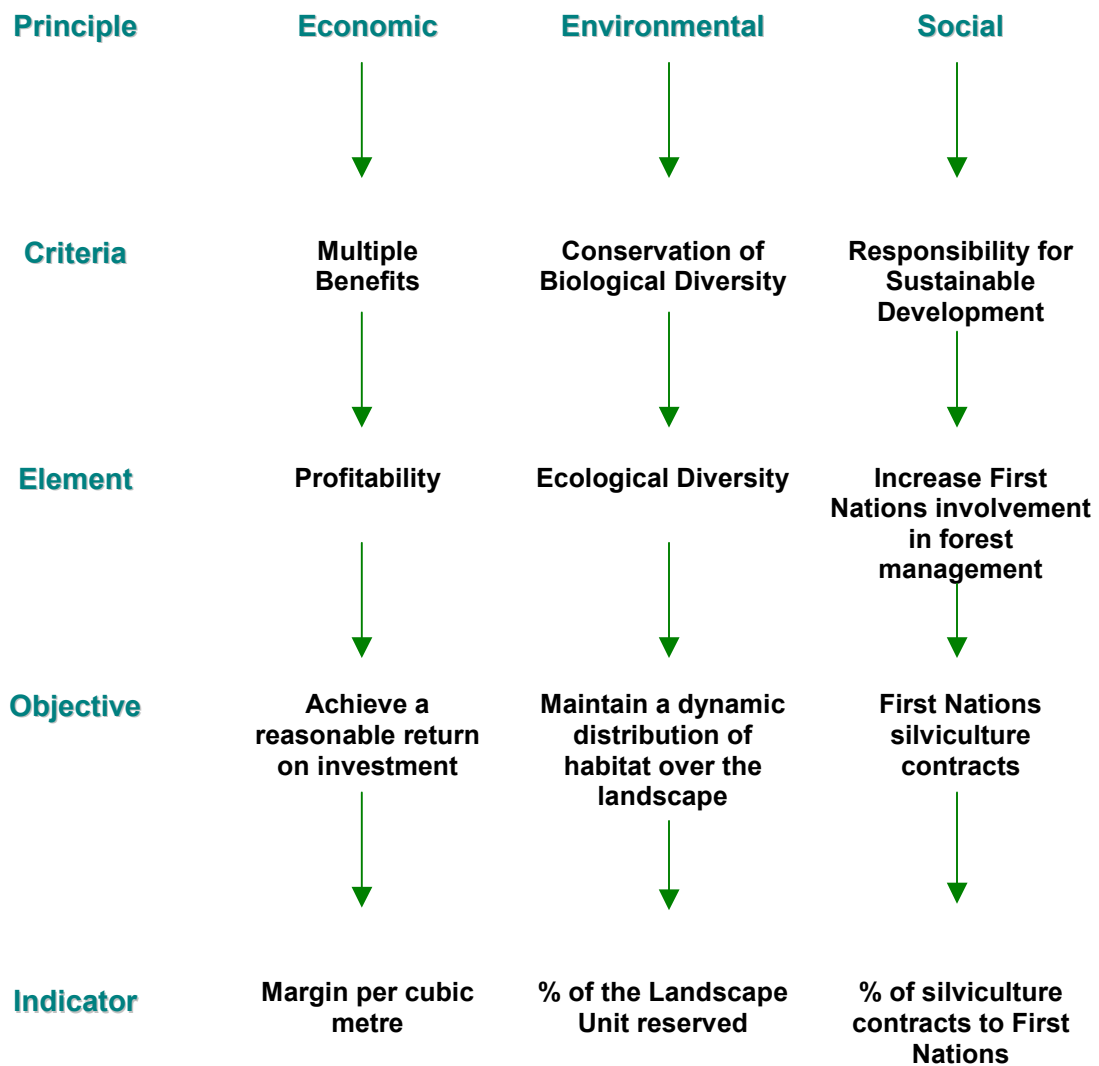
establishing a program for monitoring rare, threatened and endangered species

- Maintaining an active salmonid enhancement program to sustain or increase salmonid populations through hatchery and habitat restoration programs.
- Implementing restoration programs to address issues arising from past practices.
- Promoting a representative protected areas system within and adjacent to company forest tenures.
- Ensuring the economic health and sustainability of associated, resource-dependent communities.
- Involving the public and stakeholders in meaningful consultation on all aspects of our forest management.
- Encouraging public participation in identifying and assessing sustainable forest management goals, objectives and measurable indicators.
- Cooperating and consulting with community resource boards and stakeholder groups with resource management mandates.
- Managing operations to safeguard the health and safety of employees, contractors and the public.
- Developing our human resources through continuous training and recognition of employee contributions
- Maintaining and enhancing opportunities for public enjoyment of forests through an open road policy as well as recreation site development and management.
- Respecting and recognizing First Nations traditional territories through consultation and involvement in forest development planning
- Providing for First Nations' participation in setting and achieving sustainable forest management goals, objectives and measurable indicators.
- Establishing employment opportunities through cooperative ventures with First Nations to increase the benefits of our operations to their economic and social well being
- Supporting the fair and affordable settlement of aboriginal land claims through negotiated treaties with involvement of all stakeholders
- Achieving a fair return to shareholders by operating in a financially successful manner
- Promptly regenerating all areas after harvest with ecologically appropriate species to maintain and enhance forest growth
- Undertaking silviculture practices to enhance the value and volume of young forest stands.
- We will maintain and further our competitive position in the global forest product marketplace based on our long-standing commitment to innovation, research, leadership and sustainable forest management.

2.2 Linking WFP's Sustainable Forest Management Principles to Criteria and Indicators

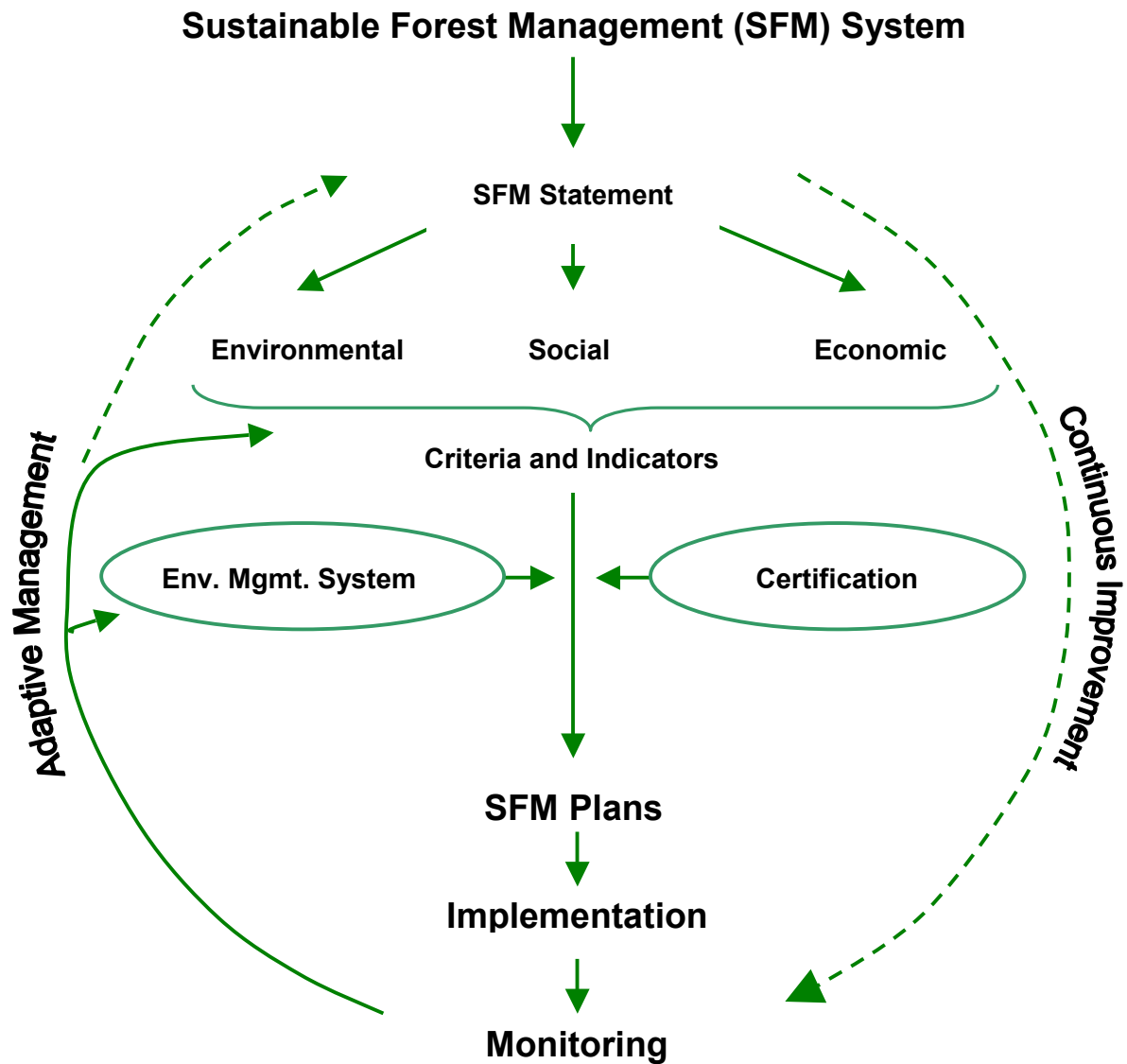
2.2.1 Hierarchy of SFM Levels

WFP's forest tenure Management Plans have been developed to implement in an effective manner the company's sustainable forest management (SFM) program and ensure compatibility with the various forest certification schemes available to the licensee. The plan focuses on three main principles followed by a series of increasingly specific Criteria, Elements, Objectives and Indicators. The following chart illustrates the concept using specific examples that are used to monitor the effectiveness of meeting identified Principles, Criteria and Indicators.



2.2.2 Sustainable Forest Management SFM system

Sustainable Forest Management is WFP's commitment, as stated in its policies, management plans and supported by our Environmental Management System and forest certification efforts. Criteria and Indicators are identified and monitored through audits and annual reporting. Results drive an adaptive management approach based on continuous improvement. The following illustration below outlines the systems.



2.3 WFP's Forest Certification Initiatives

The stewardship principles are essential components in the Company's objective to be certified under the Forest Stewardship Council Principles and Criteria (FSC), Canada's National Sustainable Forest Management System Standard, CAN/CSA-Z809-96 (CSA) and ISO 14001 Environmental Management System Standard (ISO). Linkages between WFP's sustainable forest management approach and the principles, criteria and requirements of FSC, CSA and ISO are embodied in this Management Plan and articulated in the Plan objectives, strategies and performance targets.

Measurable performance targets are set and monitored as a means of meeting the Plan objectives. Some targets will be established and reviewed annually while others will be set and reviewed periodically. TFL 6 operations were certified to ISO 14001 standards in April 2000 by QMI, an ISO certifier. Appendix XVI provides details of the ISO 14001 program. TFL 6 operations along with other North Vancouver Island Region tenures were assessed by SGS Forestry, an accredited certifier, against the Forest Stewardship Council's Principles and Criteria earlier in 2000. A certification decision has not been reached at this time, and work continues in addressing correction action required that are outstanding from the 2000 assessment work. As part of the CSA process in January 2001 a Community Advisory Group was formed to help define local values, goals, indicators and objectives for TFL 6 and adjacent company tenures. As the work of the group is still in its infancy and values, goals, indicators and objectives have not been developed yet. It is expected that the information attained from this local advisory group will be incorporated into the Management Plan as an amendment along with changes required as a result of further FSC related requirements. Depending on the level of FSC and CSA changes required, the company may consider a separate plan as these initiatives cover both TFL 6 and other adjacent tenures.

WFP will report on performance in achieving targets and Plan objectives. The TFL annual report will be one of the main reporting mechanisms used. Others include corporate reports to shareholders and audits (both external and internal).

3.0 FOREST MANAGEMENT INVENTORY AND PLANNING

3.1 Resource Inventories

Resource Inventories are maintained for the TFL land base for: forest cover, environmentally sensitive areas (ESAs), operability, recreation features (including caves and karst), visual landscape, terrain stability, ownership, archaeological resource potential, ecosystems, streams, wildlife, and potential High Conservation Value Forests attributes. Table 2 summarises the status of each inventory.

Table 2 – Resource Inventory Status

Item	Status	Within GIS	MoF Acceptance Date	Plan
Timber Inventory	Completed in 1970. MoF planned audit deferred pending VRI.	Yes	Yes -Site Index by Nussbaum 18-April-97	Inventory revisions updated annually. Vegetation Resource Inventory commenced in 1999.
Ecosystems	Completed Lewis 1982-85. Minor revisions in 1998	Yes	Used in MP 8	
Terrain Stability	Completed in 1995 and 1998 by Terence Lewis et al.	Yes	Yes 12-Apr-99	
ESAs	No longer used for planning. New inventories replace the need for this classification.	No		
Recreation Inventory	Completed in 1990 by Neufeld Reeve MacDougal Consultants	Yes	Yes 2-July-91	Update to new standards to be completed in 2001.
Recreation Analysis (ROS)	Completed in 1990 by Neufeld Reeve MacDougal Consultants	Yes	Yes 2-July-91	Update to new standards to be completed in 2001.
Visual Landscape Inventory	Completed in 1994 by Neufeld Reeve MacDougal Consultants	Yes	Yes 18-Oct-95	Visual Mitigation strategy conducted in 1998. Upgrading of Rec. and Visual Inventory to new standards currently being done by MoF as part of Regional Roll-up.
Wildlife Habitat	Ongoing			UWR surveys, eagle nest surveys, habitat modeling for Marbled Murrelet, deer, elk, and bear.
Stream Classification	FDP process has updated to FPC Riparian Classes for a large part of the TFL - 1998	Yes		Stream Inventories will continue to be updates through further operational field sampling.
Karst Inventory	Overview Completed 1979 – Paul Griffith			Inventory not released at the request of the BC Speleological Society. Although, a 1:250,000 reconnaissance level inventory is available for forest development planning.
Archaeological	Archaeological Overview Assessment completed by I. R. Wilson of Port McNeill Forest District in 1995.	Provincial sites only		Presently in discussions with Quatsino First Nation to access less sensitive information from their traditional use survey.
Operability	Completed by WFP in 1999.	Yes	Internal MoF Memo 21-Feb-99	
20-Year Plan	Timber supply model spatially explicit	Yes		20-Year Projection of modeled harvest blocks submitted in preparation for MP 9.

3.1.1 Forest Cover

The most recent field inventory was done between 1968 and 1970. Subsequently, this inventory has been updated to account for the area harvested and regenerated as well as other changes. This updated inventory was used for MP 8. Because of its age, it has no formal letter of approval. However at the time it was done, methodology followed MoF policies and procedures.

WFP has recently begun a new Vegetation Resource Inventory (VRI) project for TFL 6 that is planned for completion in 2001. Funding for the VRI will come from WFP's 5-year agreement with Forest Renewal BC. Preliminary discussions with the MoF Region have been undertaken (May 11, 1999). Photo interpretation and quick plot ground samples were completed in 1999. A more intensive ground survey program will be completed in 2000.

3.1.2 Environmentally Sensitive Areas

Areas assessed as sensitive or valuable for other resource values have been delineated through inventories conducted before and since MP 8. Management objectives reflecting the presence of these areas are in other sections of this document, including terrain stability and soil sensitivity, wildlife habitat, riparian and recreational reserves.

3.1.3 Operability

Recently 1:20,000 scale operability mapping has been updated for the TFL. The terms of reference for the new operability mapping were submitted to the Ministry of Forests in February 1999 and approved by the Regional Manager in June 1999. A field review with district staff was undertaken in October 1999 and final adjustments were made to the mapping in November 1999. Approximately 98% of operable forest types can be harvested with conventional cable and ground based methods. Helicopter or skyline systems will be utilized to harvest the remaining 2% of the operable landbase. Appendix XV illustrates the first 20 years of block sequencing used by the timber supply model to generate the "current management" harvest flow in the timber supply analysis (Appendix III).

3.1.4 Recreation

The current recreation features inventory was completed in 1993. The inventory consists 1:50,000 map sheets showing recreation feature classes, and a report describing the recreation and biophysical features and a management classification. This inventory has classified and mapped the licence area into five Recreation Opportunity Spectrum Classes based on access, feature significance and recreational use.

In 1995, WFP submitted a subzoning proposal to the Land Use Co-ordination Office for Special Management Zone 2. Part of the proposal outlined a West Coast recreation and biodiversity reserve from Grant Bay to Cape Scott Park. In 1996 and 1997 the outer West Coast was evaluated for a potential trail route between Grant Bay and Cape Scott. The evaluation reaffirmed earlier work by the British Schools Exploration Society, to locate a trail route connecting recreational and scenic values in the area but that there would be issues with the trail route depending on the shape that recreational development would take. In 1997/98, the route was further evaluated by an engineering consultant and a number of challenges (stream and river crossings, treacherous terrain, inaccessible beaches on high and storm tides, etc.) were identified. Due to the significant construction cost estimates, the program was not deemed feasible. Also, the Cape Scott and proposed North Coast trails provide ample amounts of remote hiking and other more accessible routes fill an important

demand niche. WFP has therefore changed its philosophy on the recreation vision for the area. Future development will focus on providing high quality access routes to less hazardous coastal features.

An analysis of the inventory was completed in 1999 for recreational planning over the life of MP 8. As there were no significant changes to the recreational use in the TFL, a re-inventory was not deemed necessary. Although updates to the Recreation Feature Inventory and the Recreation Opportunity Spectrum are targeted to be completed to the new standards within 2001.

Since 1995, two new recreation sites and trails have been constructed in TFL 6. Significant upgrades have been made to eleven recreation sites and trails. Total spending on recreation sites since 1995 has exceeded \$405,000. Table 3 lists new and upgraded recreational sites and trails completed from 1995 - 1999. Figure 6 shows the location of the major recreation sites. Appendix XI provides detailed sketches of developed facilities. Appendix XII contains a copy of the North Vancouver Island Region Visitors Guide.

Table 3 – Recreational Sites and Trails in and around TFL 6: 1995 – 1999

<u>New Recreation Sites and Trails</u>	<u>Estimated Cost (\$)</u>
• Hecht Beach Trail	35,000
• Palmerston Recreation Area – toilets, campsites	25,000
<u>Upgrades to existing Recreation Sites and Trails</u>	
• Beaver Lake Recreation Area – tables, firewood shelter	7,500
• Beaver Lake Trail – trail extension	30,000
• Lady Ellen Trail – trail maintenance	2,000
• Mahatta River Campsite – firewood shelter	1,500
• Marble River Campsite – toilets, firewood shelters, Tables	10,000
• O'Connell Lake Recreation Area – toilet, fire rings, Campsites	3,000
• O'Connor Lake Recreation Area – tables, firewood Shelter, toilets	20,000
• Rupert Arm Recreation Area – new shelter	20,000
• San Josef Recreation Area – firewood shelter, toilets	15,000
• Spruce Bay Recreation Area – tables, toilets	12,000
• Spruce Bay Old Growth Trail – trail extension	31,000

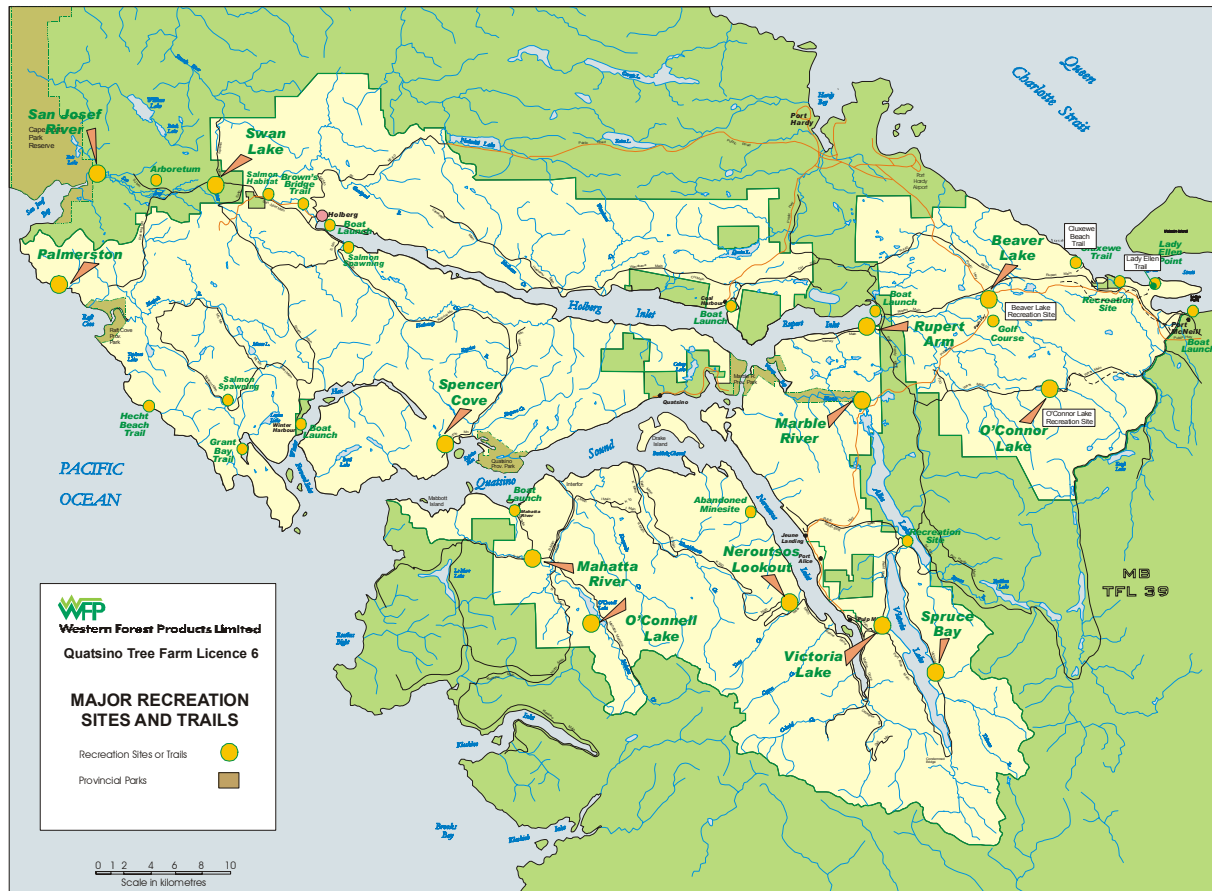


Figure 6 – Major Recreation Sites and Trails

3.1.5 Visual Landscape

The current visual landscape inventory was completed in 1992. Visual Quality Objective Mitigation Strategy was completed in 1998. In 1999 designated Scenic Areas within the TFL were made known by the MoF. Landscape sensitivity and interim visual quality classes for management have been assigned within the scenic area to manage the visual values. Generally the scenic area is made up of the land base visible from main road and marine travel corridors. Visual design techniques and alternative silviculture systems are used in the design of cutblocks in visually sensitive areas.

Upgrading of Visual Landscape Inventory to new standards is currently being done by MoF as part of the Regional Roll-up.

3.1.6 Terrain Stability

Terrain Stability Mapping (1:20,000 scale) was completed for the TFL in December 1996 by Dr. Terry Lewis. This terrain mapping has been used to identify areas where further geotechnical analyses are needed to ensure appropriate management strategies are implemented for soil conservation. This mapping also provides guidance during the TFL operability revisions. Only under rare circumstances is road construction proposed in terrain Class V areas. On terrain Class IV (and V) areas, roads and harvesting will occur only after further geotechnical analyses and careful planning including the review and approval by government review agencies takes place.

3.1.7 Ownership and other Tenures

Ownership and land status have been updated for the TFL 6 area. Since MP 8 three new Class A provincial parks have been created within the TFL. They are the Quatsino Provincial Park (654 ha), Marble Provincial Park (1,512 ha), and the Misty Lake Provincial Park (55 ha).

3.1.8 Archaeological Resource Potential

An Archaeological Overview Assessment (AOA) for the Port McNeill Forest District including TFL 6 was completed in 1995. This overview identifies archaeological sites and resources and indicates where past human activities are likely to have occurred. The AOA is used in planning. Areas of high potential are subject to field reconnaissance and inventory followed by further analyses to ensure the protection of special features. The Quatsino First Nation completed a Traditional Use Study for their traditional territory in 1998. WFP is working on a sharing agreement for the less sensitive information in the study. A Traditional Use Study has been completed for the traditional territories of the Kwakiutl and Tlatlasikwala First Nations as part of the Galgalis Traditional Use Study.

3.1.9 Ecosystem Mapping

An ecological inventory and mapping project has been completed by Dr. T. Lewis. The system has been used as a basis for forest management activities and site productivity since 1985. Table 4 provides a summary of the Ecosystems that Lewis has identified within TFL 6.

Table 4 – TFL 6 Ecosystem Classification System

Ecosystems of the Submontane	
S1	Zonal Salal – Moss
S1ch	- Red Cedar – Western Hemlock Phase
S1ha	- Western Hemlock – Amabilis Fir Phase
S2	Red Cedar – Salal, Folisol over rock
S2F	- Douglas Fir Phase
S2P	- Pine Phase
S3	Alluvial Sitka Spruce – Sword Fern – Foam Flower
S3B	- Beach Phase
S4	Alluvial Red Alder – Sitka Spruce – Herb
S5	Wet Alluvial Sitka Spruce – Red Cedar – Skunk Cabbage
S6	Red Cedar – Skunk Cabbage – Sphagnum
S7	Shore Pine – Red Cedar Swamp Forest
S8	Shore Pine Bog
S9	Treeless Sphagnum Bog
S10	Red Cedar – Western Hemlock – Salal, Folisol over water table
S11	Sweet Gale – Hardhack – Sedge Fens
S12	Limestone Holly Fern
S12ch	- Red Cedar – Western Hemlock Phase
S12F	- Douglas fir Phase
S13	Colluvial Salmonberry – Sword Fern
S13L	- Limestone Phase
CP	Yellow Cedar – Shore Pine Scrub Forest
Ecosystems of the Montane	
M1	Montane Moss
M1C	- Colluvial Phase
M2	Yellow Cedar – Western Hemlock – Salal
M3	Oakfern
M4	Wet Montane Skunk Cabbage – Marsh Marrigold
M5	Montane Devil's Club
Sub-Alpine and Alpine Ecosystems	
MH1	Closed Zonal Mountain Hemlock Forest
MH1C	- Colluvial Phase
MH2	Mountain Hemlock – Rock Complex
P	Mountain Hemlock Parkland
A	Alpine
AT	Avalanche Track Ecosystems

The use of the ecological information for estimation of site productivity was raised as an issue in the approval of MP 8. A project was undertaken by J. S. Thrower in 1996 to assess the validity of this approach. The report confirmed the approach used in MP 8. The average site index Thrower obtained was 0.3 m higher than that used in MP 8. A letter approving the Thrower report was received from A. Nussbaum of Research Branch on April 18, 1997. This letter confirmed that this approach was suitable for use in MP 9.

3.1.10 Streams

A comprehensive riparian inventory has not been completed for the entire TFL. However, operational stream inventories have been conducted since 1988 and a Resource Inventory Committee project is underway. This mapping project will provide information on fish distribution, fish habitat and habitat restoration opportunities. These two sources will provide a great deal of site specific survey information for operational use. WFP intends to utilize this detailed work in conjunction with GIS modelling to obtain an overall estimate of riparian classes for the TFL. Forest Practices Code riparian reserves will then be applied to all known and predicted fish bearing streams. This approach is outlined in detail in the Timber Supply Analysis information package.

3.1.11 Wildlife

Until recently wildlife inventories were focused on specific species. In the late 1970's and early 1980's fourteen deer winter range areas totalling 2,040 ha were established in TFL 6 and removed from the timber harvesting land base. Under the Forest Practices Code of BC Act, Operational Planning Regulation approximately 60% (1,031 ha) of these sites were grandparented (as of October 1998) as ungulate winter ranges (UWR). Objectives for managing ungulate winter ranges have been set by the Forest District Manager and designated environment official. By October 2003 boundaries for ungulate winter ranges must be made permanent. WFP will work

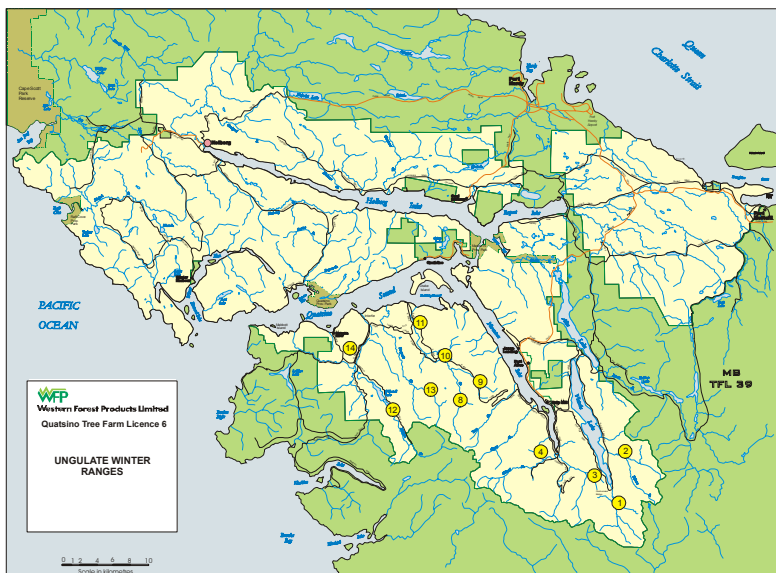


Figure 7 – Ungulate Winter Ranges

with both the MoF and MOELP to finalize ungulate winter range boundaries and develop management strategies for them. In 1998 and 1999 the fourteen deer winter ranges mentioned above as well as other potential wildlife habitat areas were evaluated to identify and classify potential habitat for ungulates, Marbled Murrelets as well as several other species. Although many of the non-designated UWRs were shown to have low value for deer they do contain some medium to higher value Marbled Murrelet habitat. These areas are located in the Koprino River area and they or equivalent areas may be designated as Wildlife Habitat Areas or Old Growth Management Areas that focus on Marbled Murrelet. Figure 7 shows the location of ungulate winter ranges.

As interest in wildlife broadened, so did inventory efforts. D. Blood & Associates completed a TFL 6 biodiversity overview in 1996 that has been used for management purposes. The overview provides a comprehensive inventory of flora and fauna that are likely found within the licence area. The overview identified four issues that need to be addressed to improve the long-term biodiversity of the area.

1. Meeting landscape goals for old seral representativeness and protecting under represented plant communities,
2. Maintained or recruiting habitat for species of conservation interest (red and blue listed or regionally significant),
3. Protecting aquatic habitat for vulnerable fish stocks, and
4. Meeting landscape goals for connectivity through Forest Ecosystem Networks (FEN).

Issues 1 through 3 are addressed through current government forest policy and the management of High Conservation Value Forests attributes as outlined within Appendix XX. Current researchers suggest that connectivity in coastal forest may be of lesser importance than originally thought but specific management strategies will be considered as Landscape Unit Planning is implemented. Blood also identified several measures required to protect fish and wildlife habitats within the licence area. These measures were considered and addressed by Kremsater in the "Strategy Outline for Managing Wildlife" as part of the High Conservation Value Forests report. A listing of wildlife species and plant communities found in the overview are found in Appendix XX.

In 2000, inventory work for Queen Charlotte (northern) goshawks (*Accipiter gentilis laingi*) was initiated in conjunction with several other adjacent tenures. Queen Charlotte goshawks are red listed and subject to protection measures under the Identified Wildlife Management Strategy. To date, the preliminary work has identified one nest located near Nordstrom Point within the TFL. As no Wildlife Habitat Areas have been identified within the TFL, protection measures will be established in consultation with MOELP, in areas where cutblocks are proposed near known nest sites.

Field inventory work has started for Marbled Murrelets to validate a habitat suitability model developed in 1999. In 2000, detection surveys were conducted to determine the birds relative use of identified habitat. The results of this work are expected to be available in 2001.

An inventory of bald eagle nests is maintained for operational planning purposes. A total of 177 eagle nests have been identified in the TFL.

3.1.12 High Conservation Value Forests

High Conservation Value Forests are defined as those forests that possess one or more of the following attributes:

- Forest areas containing globally, regionally or nationally significant:
 - Concentrations of biodiversity values (e.g. endemism, endangered species refugia); and/or
 - Large landscape level forests, contained within, or containing the management unit, where viable populations of most if not all naturally occurring species exist in natural patterns of distribution and abundance
- Forest areas that are in or contain, rare, threatened or endangered ecosystems
- Forest areas that provide basic services of nature in critical situations (e.g. watershed protection, erosion control)
- Forest areas fundamental to meeting basic needs of communities (e.g. subsistence, health) and/or critical to local communities' traditional cultural identity (areas of cultural,

ecological, economic or religious significance identified in cooperation with such local communities).

While the term High Conservation Value Forests (HCVF) is new, the concept behind it is not. WFP has been engaged in various forms of public consultation over the past three decades which have identified several attributes that are considered HCVF within the Licence area. Many of the HCVF attributes identified thus far are included within the inventories discussed earlier. Other attributes that have not been specifically identified in current inventories (rare, threatened, or endangered ecosystems or plants) can generally be flagged by correlating the comprehensive ecosystem inventory completed by Lewis with the information provided by the B. C. Conservation Data Centre.

During the term of the plan, WFP will identify and locate known HCVF attributes and develop management strategies to provide for these attributes. The management strategies will be incorporated into Landscape Unit and operational planning processes. It is likely the management strategies will go beyond the current planning policies and processes guided by the company's scientific advisors. It is expected that the process of developing local definitions, mapping and management strategies will be ongoing and subject to continual evolution as new information and new techniques are learned.

3.2 Planning Linkages - Strategic and Operational Planning

The hierarchy of plans governing activities on TFL 6 is as follows: Vancouver Island Summary Land Use Plan (VISLUP) ⇒ Landscape Unit (LU) Planning ⇒ Management Plan 9, TFL 6 (and assumed Timber Supply Analysis) ⇒ Operational Plans (Forest Development Plans, Silviculture Prescriptions etc). Strategies and objectives both within and between these various plans must be consistent. Generally, Higher Level Plans such as the VISLUP and LU Plans allow for legally enforceable objectives to be established under the Forest Practices Code. These Higher Level Plan objectives give direction to subordinate plans that contain the detail for implementation.

Linkages to WFP's sustainable forest management (SFM) principles are made with both strategic level and operational level plans. SFM principles incorporate the Forest Practices Code and Regulations, Forest Act and other related legislation as minimum standards.

3.2.1 Regional Plans

The Vancouver Island Summary Land Use Plan is a Regional Plan, which covers the TFL area. Eight Resource Management Zones (RMZs) have been established for the TFL landbase. Of the eight zones, two are classified as Special Management Zones (SMZs), two are classified as General Management Zones (GMZs), and four are classified as Enhanced Management Zones (EMZs). General management regimes for timber, biodiversity, cave karst, water, visual quality, cultural heritage, fish, recreation, access, wildlife and tourism have been established for each zone. These regimes are based on the requirements for integrated resource management and reflect pertinent resource legislation and policy.

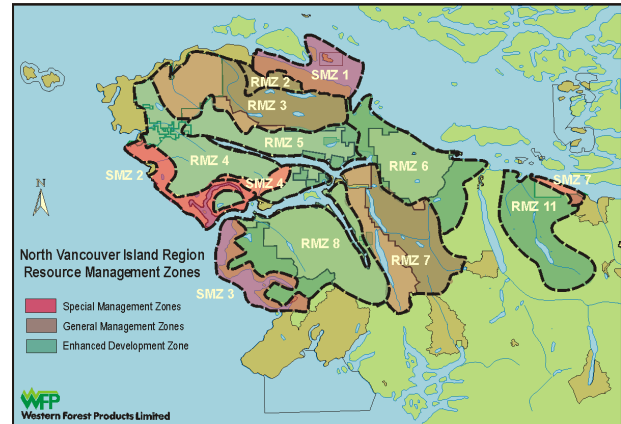


Figure 8 – Resource Management Zones of TFL 6

Table 5 summarizes the RMZs covering the TFL.

Table 5 – Resource Management Zones and Regimes

RMZ #	RMZ Name	TIM	WAT	FISH	WL	BIO	VIS	REC	TOU	C/K	CULT HER	ACC
SMZ2	West Coast Nahwitti Lowlands	S	G	G	S	G	S	S	S	G	G	G
SMZ4	Koprino	S	G	G	S	G	G	G	G	G	G	G
3	Nahwitti - Tsulquate	G	CW/G	G	G	G	G	G	G	G	G	G
4	San Josef – Koprino	E	G	G	G	G	G	G	G	G	G	G
5	Holberg	E	CW/G	G	G	B	G	G	G	G	G	G
6	Keogh Cluxewe	ES/ETH		G	G	B	G	G	G	G	G	G
7	Marble	G/ES	G	G	G	G	G	G	G	G	G	G
8	Mahatta – Neroutsos	E	G	G	G	B	G	G	G	G	G	G

Note: Cave/Karst management is limited to those RMZs with known features or high probability of such occurrence.

TIM: Timber; BIO Biodiversity; C/K Cave Karst; WAT Water; VIS Visual; CULT.HER Cultural Heritage; FISH Fish; REC Recreation; ACC Access; WL Wildlife; TOU Tourism;

Regimes: S–Special; G–General; B–Basic; H–Higher; E–Enhanced; ES–Enhanced Silviculture; ETH–Enhanced Timber Harvesting; CW–Community Water

Some objectives have been made Higher Level Plans under the Forest Practices Code. These objectives affect operational planning in SMZs and EMZs. SMZs Higher Level Plan objectives deal with cut block structure, size and silviculture systems, joint approval of Forest Development Plans and visual quality management. EMZs Higher Level Plan objectives deal with Landscape Unit planning, cut block size and green-up, hydrology and reforestation species selection.

3.2.2 Landscape Unit Plans

In March 1999, the provincial government released the Landscape Unit Planning guide. Also in early 1999 a Regional Landscape Unit Planning Strategy was developed for the Vancouver Region, which focuses on the identification and establishment of old growth management areas (OGMAs) to meet biodiversity objectives. The strategy also deals with the establishment of wildlife tree patches (WTPs) and the protection of marbled murrelet habitat. Objectives for other resources can also be developed through LU Planning but are not mandatory.

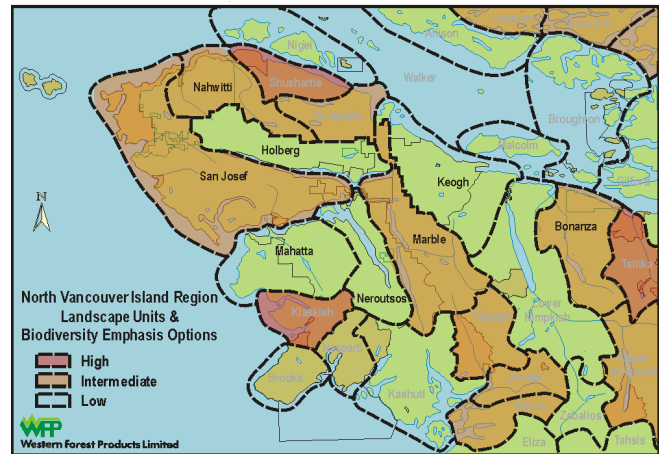


Figure 9 – Landscape Units of TFL 6

Six landscape units cover the TFL landbase. The boundaries of the LUs generally follow those of the RMZs. A high, intermediate or low Biodiversity Emphasis Option (BEO) has been assigned to each LU based on a combination of factors including the amount of protected area within the LU, percent of old forest, ecosystem complexity, sensitivity to development, connectivity, forest productivity and timber operability. WFP has begun the delineation of OGMAs in some LUs in concert with MoF and MOELP. The completion of this process should take place in 2001. WFP will continue to take a lead role in landscape unit planning for the area covered by the Plan.

Table 6 shows the LUs and BEOs for TFL 6.

Table 6 – LUs and BEOs for TFL Landbase

LU Name	BEO	Total Area in TFL 6 (ha)	Productive Area (ha)
Holberg	Low	33,500	31,859
Keogh	Low	31,337	29,856
Mahatta	Low	24,440	23,444
Marble	Intermediate	23,649	18,347
Neroutsos	Low	26,523	23,463
San Josef	Intermediate	61,607	58,522

Under WFP's environmental principle and the conservation of biological diversity, criteria and indicators have been established to monitor the amount of the landscape unit reserved from harvest which would include OGMAs, WTPs and other reserves.

3.2.3 Forest Development Plans

Forest Development Plans (FDPs) are required under the Forest Practices Code of BC Act and the Operational Planning Regulation to direct forest management activities and practices. Specifically FDPs show the location and orderly development of proposed harvesting and road construction, maintenance and deactivation. The FDP also includes information on access management and the maintenance and protection of other resource values in the area

covered by the plan. The FDP must be consistent with Higher Level Plans and made available for review by resource agencies, First Nations and the public before approval is considered by the Ministry of Forests. FDPs are submitted either annually or every two years and contain enough blocks to equal at least five years of annual harvest. Several indicators related to WFP's SFM principles have been established which will be reported on in the Forest Development Plan. The list includes:

- ha maintained as stand level reserves
- # of accidental fires/ha burned
- ha harvested by silviculture system
- harvest system by volume
- average opening size
- km of road maintained
- km of road deactivated
- ha of coastal watershed assessment procedures (CWAPs) completed by year
- records of meetings and correspondence with First Nations
- # of archaeological impact assessments (AIAs)
- # and types of contacts/meetings/consultations with communities.

3.2.4 Stand Level Prescriptions

A number of different plans and prescriptions detailing forest management activities are prepared for review and approval by the MoF and other government agencies. Silviculture Prescriptions, Stand Management Prescriptions, Access Management Plans and Logging Plans are referred to as operational plans under the Forest Practices Code of BC Act, Operational Planning Regulation. Deactivation prescriptions detail road deactivation activities and are regulated by the FPC, Forest Road Regulation. Appendix V includes an example of a Silviculture Prescription.

Cutting and road construction authority is given by MoF through Cutting Permits and Road Permits. These permits are granted under the Forest Act.

Much of the content requirements for these plans, prescriptions and permit applications are defined by legislation. Policy guidelines and guidebooks also give direction in the development of plans, prescriptions and permit applications.

3.2.5 Adaptive Management

Forestry plans must recognize limited knowledge and uncertainty. For this reason, ecosystem management plans are based upon adaptive management principles. These include the maintenance of ecosystems and habitats across a landscape, consideration of land management impacts at both the stand and landscape levels and striking a balance between timber and other resources. As further knowledge is gained, management practices will adapt and utilize the knowledge in the development of future management strategies.

3.2.6 Ecosystem Based Management

An ecosystem-based approach to planning and operations that protects cultural, social and ecological values will be implemented. The basis of this approach is a comprehensive ecological classification (Section 3.1.9) that has been in place for almost 20 years in TFL 6, and guides all forest management activities in the tenure. The classification system is very closely tied to the site ecology, sensitivity and productivity. Detailed site specific prescriptions are developed incorporating this information and all other resource values. Management activities are implemented based on these prescriptions.

4.0 SUSTAINABLE FOREST MANAGEMENT STRATEGY

4.1 Economically Viable Forest Management

Principle

4.1.1 Multiple Benefits

Criteria

4.1.1.1 Timber Supply

Element

Objective	Indicator
Harvest AAC	% achievement of AAC, 5 year cut control estimated total value of timber produced
Prevent timber loss	ha of non-recoverable losses due to wind, fire
Efficient utilization	m ³ of billable waste

WFP is committed to economically harvesting the licensee portion of the total proposed TFL 6 AAC of 1,469,900 m³ in accordance with its SFM principles and the FPC (Appendix III). The licensee portion of the proposed AAC is 1,456,923 m³. The Small Business Forest Enterprise Program AAC is 13,242 m³. The schedule "B" pro-rate is 0.777. Table 7 shows the actual harvest to AAC performance since the granting of the TFL in 1950.

Table 7 – Cut Control Performance 1950-1999

Period	Allowable Cut Available To Licensee	Chargeable Cut
1951-1960	4,168,156	4,242,856
1961-1965	3,652,867	3,937,601
1966-1970	5,703,003	5,749,372
1971-1972	2,619,304	1,862,800
1973-1977	6,250,735	5,915,767
1978-1979	2,361,622	2,544,317
1980-1984	6,321,622	6,618,305
1985-1989	6,462,600	6,944,146
1990-1994	5,906,900	6,941,356
1995-1999 ²	6,536,720	6,233,305
Total	49,983,530	49,999,825
² Includes Chargeable Cut before carry-forward adjustments to end of 1999.		

Detailed assumptions used for MP 9 Timber Supply Review can be found in the TFL 6 Timber Supply Analysis package, Appendix II.

Harvesting will emphasize the maximum recovery of saw and pulp log grades. Log recovery will conform to the Vancouver Forest Region Close Utilization Standards as a minimum. These standards are:

	Old Growth	Second Growth
Diameter breast height (cm)	17.5	12.5
Maximum stump height (cm)	30	30
Minimum top diameter (cm)	15	10
Minimum log length (m)	3	3

The company may exceed these minimum standards by recovering low-grade pulp quality logs. Generally, these logs are marginal Y grade specifications whose utilization is optional. Utilization is also kept high by the salvage of cedar shake and shingle blocks and to a lesser extent cypress cants and spruce guitar blocks from logged blocks.

Utilization levels will be monitored by conducting residue surveys to determine the volume of avoidable and unavoidable fibre left on logged blocks. Current MoF policy for avoidable waste billing is 35 m³/ha for old growth and 10 m³/ha for second growth. WFP's goal is to be below these targets. Waste survey results are submitted to the MoF. The MoF enters avoidable and unavoidable waste into the cut control record. Where CWD requirements dictate, some previously "avoidable" waste may need to be deemed "unavoidable".

Merchantable deciduous (alder) volumes are minimal and should not be included in the AAC. The company will consider sales of deciduous volumes and conversion of alder sites if the opportunity arises.

Salvage from stands damaged by blowdown, fire, pest or disease infestations will be given a high priority. Assessments of these stands will include the extent and degree of damage, feasibility for harvest and relevant value with respect to biological diversity.

As part of WFP's SFM principles, three objectives have been established regarding timber supply. These are:

- harvest the AAC
- prevent timber loss
- efficient utilization

Indicators have also been established to measure our success in achieving these objectives. The indicators for the AAC are % compliance with the cut control regulation at the end of each cut control period and estimated total value of timber produced. The indicator for timber loss is to keep the number of ha of non-recoverable loss due to wind and fire to 30 ha or less on an annual basis. An internal tracking system has been implemented to monitor these issues. With respect to billable waste our indicator/objective is to have an average less than 20 m³/ha. These indicators will be monitored yearly in the TFL Annual Report.

The Timber Supply Analysis has been completed for review by the Chief Forester and proposes, after allowances for non-recoverable losses, that the harvest level for Tree Farm Licence 6 be reduced to 1,469,900 m³/year through the period 2001 – 2005. This includes an optional, experimental commercial thinning partition of 3,000 m³/year and a partition of 14,500 m³/year for economically marginal forest to be harvested only if and when conditions permit. The basic harvest level of 1,452,400 m³/year is consistent with a step down of harvest levels forecast five years ago during the preparation of Management Plan 8.

Complan 3.0, a spatial harvest model, was used to simulate timber supply based on

current management practices in the licence area. The proposed harvest flow was derived after ensuring sustainability of environmental values. Simulation of current management practices maintained at least 30,000 ha of old forest and 83 million m³ of growing stock throughout the 250-year simulation horizon. In response to increased emphasis on environmental values, the projected harvest levels have been and are continuing below the unrestrained timber production potential of the Tree Farm Licence.

Sensitivity analyses around the current management or “base case” simulation suggest that timber supply is sensitive to land base withdrawals, adjacency and block size policies, seral stage targets, minimum stem size at harvest, and inventory changes. Further constraints would reduce flexibility and make impacts most likely forty to fifty years into the future.

After 2060, gains from current and past silviculture efforts produce an abundance of growing stock and facilitate a predicted “fall up” in harvest levels to a long term level of 1,640,000 m³/year.

4.1.1.2 Harvest Methods

Element

Objective	Indicator
Employ appropriate harvest methods	Harvest system by volume

A variety of logging systems are being utilized in TFL 6. The main conventional systems being used are mobile grapple yarding and hoe forwarding. Mobile highlead towers and skyline systems are also used. Line skidding has been used in the past. Non-conventional harvesting is done mainly with helicopters. Manual falling and log length yarding are common to all of the above mentioned systems. Figure 10 illustrates the typical timber harvesting process in TFL 6.

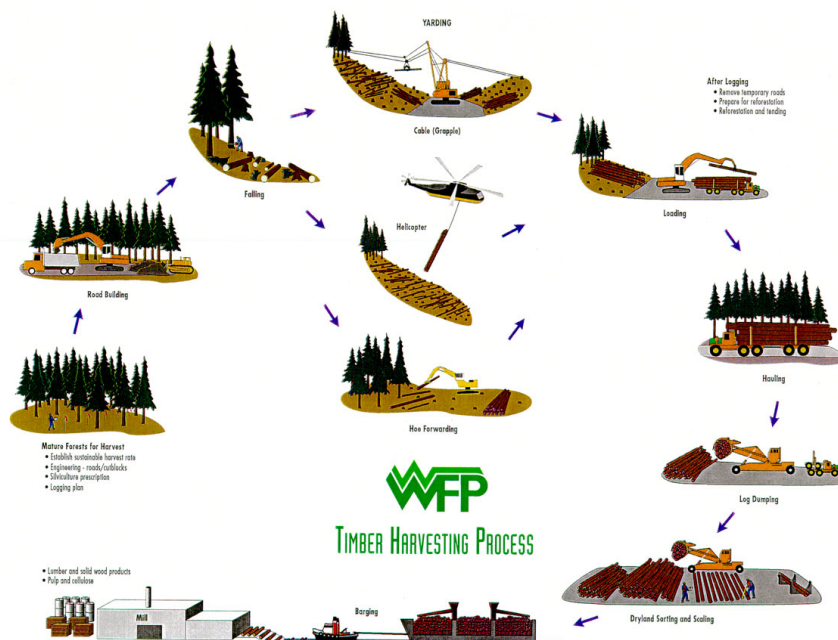


Figure 10 – Timber Harvesting Process

New operability lines have been established for the TFL delineating operable stand types and anticipated logging system. Marginal stand types such as pure hemlock balsam height class 3 stands will be harvested only when they are economically viable. Over the planning period the anticipated split between conventional and aerial logging systems will be approximately 98/2. Table 8 outlines the historic use of logging systems in TFL 6.

Table 8 – Distribution of Logging Systems

Percent of total production (approx.).

Logging Systems	Historical	1990-94	1995-1999
Conventional cable and ground based systems	100	100	98
Skyline systems*	0	0	1
Non-conventional systems: aerial and multi-span skylines	0	0	1

*Skyline system defined as a system utilizing lateral yarding and/or yarding distances over 300 meters.

WFP has experimented with mechanized falling and processing heads mounted on tracked carriages. This type of harvesting equipment may be utilized in patch clearcutting or partial cutting systems in second growth or for commercial thinning where conditions allow.

The indicator established to monitor appropriate harvest methods is the harvest system by volume. This information will be reported on an annual basis in the TFL Annual Report.

4.1.1.3 Employment

Element

Objective	Indicator
Sustain Employment Levels	Total person-days and jobs/m ³

Tree Farm Licence 6 is a significant generator of employment in logging silviculture and forest management. Other employment is created as the logs produced in TFL 6 are processed in the Port Alice Pulp Mill and other Doman mills. Table 9 summarizes the average direct employment over the past five years as a consequence of operations in TFL 6.

Table 9 – Direct Employment 1995 - 2000

Category	Average Person Days of Employment
Planning/Harvesting	81,090
Silviculture & Integrated Use	13,630
Administration	4,250
Subtotal	98,970
Manufacturing	102,725
Total	201,695

Over the planning period WFP intends to maintain employment levels through the following strategies:

- Harvest 100 % of the AAC in each cut control period
- Meet or exceed contractor requirements as per Timber Harvesting

Contract and the Subcontract Regulation

- Where possible expand the silviculture and watershed restoration programs through long term funding commitments from FRBC
- Maintain and where possible expand the minor forest products harvesting program
- Work with First Nations to expand employment opportunities

Employment strategies hinge on the degree of decline of the AAC, no diminishment with company tenure rights and continued or expanded funding from Forest Renewal BC over the next 5 years.

4.1.1.4 Contractor Commitment

Element

Objective	Indicator
Maintain contractor commitments	% of required contractor commitment achieved

Under Part 14 of the TFL 6 contract, WFP is required to ensure the harvest of timber equivalent to 50% of the harvest contributed by Crown lands be carried out by contractors according to the Timber Harvesting Contract and Sub-Contract Regulation. Over the past five years WFP has performed in excess of requirements as indicated in Table 10:

Table 10 – Compliance with Timber Harvesting Contracting Regulation

Year	% Compliance
1995	133%
1996	153%
1997	133%
1998	140%
1999	121%
2000	115%

Currently the following contractors have replaceable contracts in TFL 6:

Full Phase Logging

A.H Jackson Corp.
Koprino Logging Contractors Limited (incl. roads)
557025 BC Ltd.

Road Building

C&E Road Building Ltd.
D.D.H. Contracting Ltd.
Goodspeed Roadworks Ltd.
Holberg Contracting Ltd.
J.F. Trucking Ltd.
Juthans Trucking Ltd.
Koprino Contracting
Mike Jones Enterprises (road maintenance)
North Island Rock Pro Inc.
Poor Boy Enterprises
Port McNeill Enterprises
Raft Cove Contracting
Stan Zwicker Contracting
531088 BC Ltd.

Falling, Yard, Loading

G.L.M. Falling Limited
Leeson Lake Falling
Quatsino Forest Company
Ransom Creek Falling
W.D. Moore Logging Company (incl. roads)

Hauling

C.S.K. Hauling Ltd.
GCB Ventures Ltd.
Helgesen Logging Ltd.
Lefty's Log Hauling
MacLeod & Smith Log
MiDon Contracting
Pt. Alice Salvage Ltd.
Steele Rock Hauling Ltd.
R & K Tyson Contracting Ltd.

An annual accounting of the contracting requirements and performance is in the TFL 6 Annual Report.

There are more than 100 other contracting firms working in TFL 6 as part of forest operations. Work performed includes cruising, cut block layout, tree planting special forest products salvage and many other tasks.

4.1.1.5 Profitability

Element

Objective	Indicator
Achieve a reasonable return on investment	Margin/m ³

WFP is committed to the maintenance of economically viable forestry and manufacturing operations through the optimal utilization of timber values. Through the internal manufacture and trading of logs WFP will strive to maximize value margins and provide a competitive return on investments to our shareholders.

Return on investment and log margin average will be reported in company financial and annual reports respectively.

4.1.1.6 Non-Timber Forest Products

Element

Objective	Indicator
Encourage NTFP utilization	Volumes and type of NTFP (m ³ , kg, l)

WFP will plan for and utilize wherever possible all commercial non-timber forest products while ensuring a high standard of management and environmental control to ensure sustainability. The main non-timber forest products harvested in TFL 6 are yew bark, honey, and cedar oil.

Over the past five years roughly 7,500 kg of yew bark has been collected from cutblocks approved for falling. WFP collects \$1.00/kg to cover some of the administration costs.

The production of fireweed honey has increased over the past several years. Commercial and personal use apiaries are commonly set up in recently harvested areas throughout the TFL. WFP does not collect any administration fees for the collection of honey on lands managed by the company. However, the company does request hive location and production reports each year.

In 1998, a local First Nations entrepreneur purchased extraction equipment to produce western red cedar oil. Since the initial start up of the business the company has diversified from producing only high quality cedar oil for export into the production of cedar mulch, soap and candles. Approximately 1850 litres of oil have been extracted from cedar foliage collected from the TFL.

Chanterelles grow in many stand types throughout the TFL including pole sized second growth. Pine mushrooms are not as common as chanterelles but are found to a lesser extent within the TFL. Mushroom production has not been recorded. Salal is a common shrub growing in cutovers and open grown stands. Salal is collected from TFL lands for the floral industry but production is not currently recorded.

WFP staff are participating in various non-timber forest products forums and initiatives regarding policy development and employment opportunities. New employment opportunities will be pursued in the TFL where sufficient revenue is generated to offset administration costs, assessments and inventories indicate the activity is sustainable, and the activity will not negatively impact other forest management investments. WFP will continue to report on non-timber forest product activities in annual reports and other pertinent publications.

Designated trapping areas are illustrated in Figure 11. The level of trapping within the licence area is relatively low. WFP cooperates with the licence holders through notification, to minimize impacts on trapping resources, when development is proposed within the licence area. Maintenance of biodiversity in the TFL sustains trapping habitat.

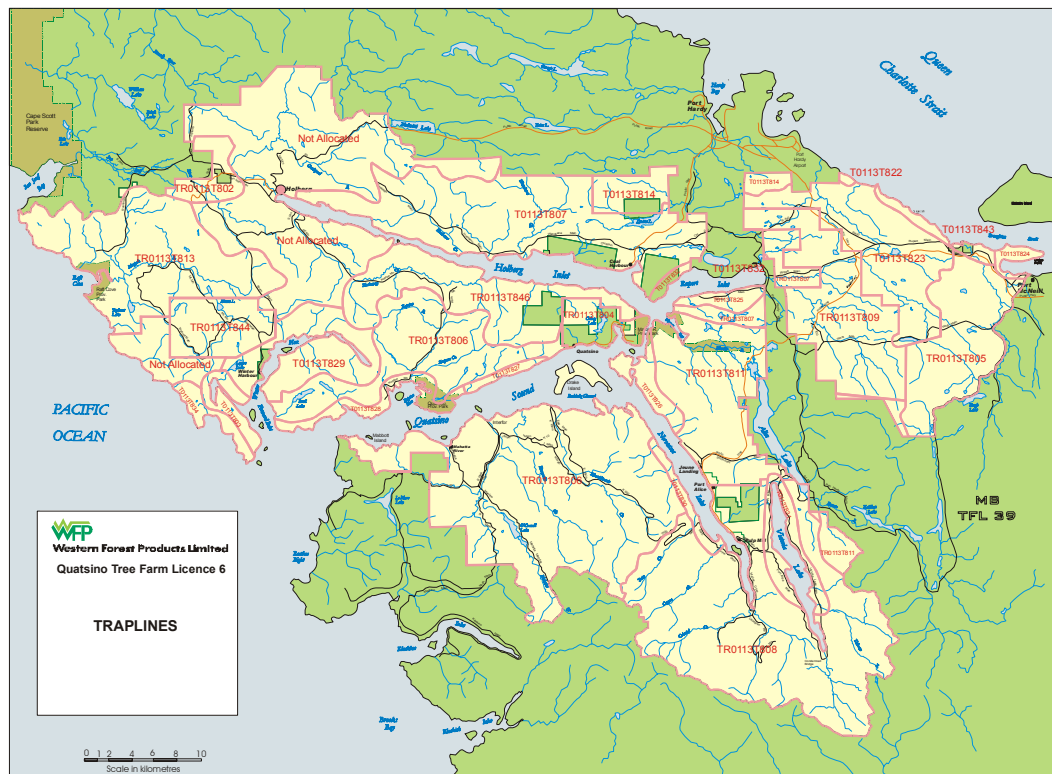


Figure 11 - Traplines

The Quatsino Sound waterways provide various opportunities for aquaculture, tourism and other commercial ventures related to the marine areas adjacent to TFL 6. Ventures of this type are important to the economic diversification and development of the North Island. WFP will cooperate with planning and development of aquatic related economic development. Figure 12 shows the location of known existing aquatic licences.

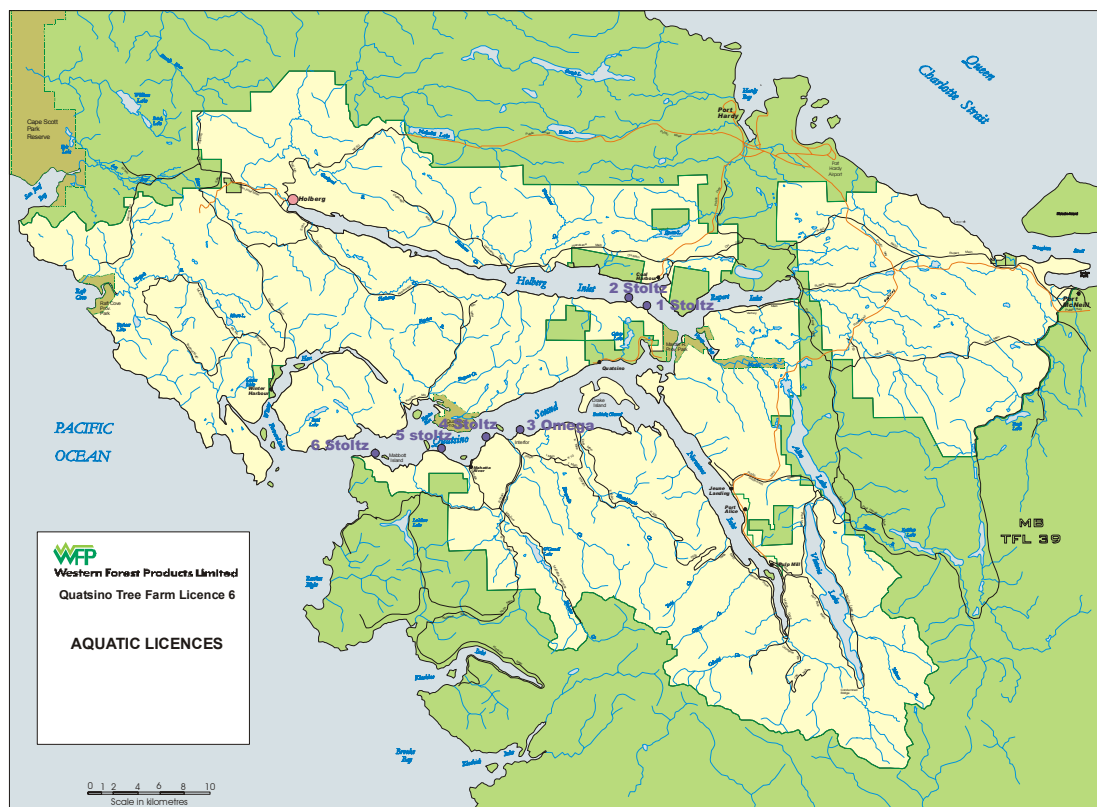


Figure 12 – Known Aquatic Licences

4.1.1.7 Minor Forest Products

Element

Objective	Indicator
Encourage utilization of minor forest products	Salvage volume
	Estimated total value of special forest products

WFP has an active minor forest products program to harvest minor forest products from its tenures within TFL 6. Established contractors salvage mainly western red cedar shake and shingle blocks in cutblocks that have been surveyed for waste and residue. Minor amounts of yellow cedar and Sitka spruce blocks and yellow cedar and Douglas fir cants are also produced. On average 10,000 - 11,000 m³ of cedar shake/shingle blocks are produced each year and 200 - 300 m³ of cants. A log salvage program began in 1995 to salvage low-grade logs left in old areas. This program produces approximately 500 – 1000 m³ per year

Culvert and buried logs from road deactivation activities produce roughly 5,000 m³ annually. This volume is hauled to dryland sorts and scaled with regular production.

WFP will continue to encourage utilization of minor forest products and will report volume produced and the estimated total value of special forest products in the annual report.

4.1.1.8 Access Management

Element

<i>Objective</i>	<i>Indicator</i>
<i>Provide for public access</i>	<i>km of road maintained</i>

A transportation system has been developed in the TFL over the past 45 years to deliver logs by truck and by water to manufacturing facilities. Roughly 2,500 km of road has been built to date of which an estimated 800 km is currently maintained.

Since 1995 roads are constructed, maintained and deactivated as per the Forest Act and the Forest Practices Code of BC Act, Forest Road Regulation.

Access management maps showing the current and planned status of roads will be prepared and made available for public comment with the Forest Development Plan.

All road users assume certain obligations for the use of roads. Road users assume the risk of accident or injury. Road users are also expected to cooperate with WFP staff in reporting and suppressing fires, conforming to company and BC Forest Service fire protection regulations, participate in Neighborhood watch programs, and report wildlife violation and criminal activities.

Road status changes will be reported annually in the Annual Report.

4.1.1.9 Recreation

Element

<i>Objective</i>	<i>Indicator</i>
<i>Maintain recreation sites and features</i>	<i>No. and type of recreation sites maintained</i> <i>User days by category</i>

Recreation management in the TFL is based on a feature inventory and special studies to evaluate future developments. The main recreational use has been for camping, hiking, fishing and hunting predominantly by local residents.

Recreation Reserves have been identified in the Grant Bay, Marble River, Spruce Bay, and Topknot Beach areas, for several years. The reserves have been withdrawn from the timber harvesting land base signifying the high recreational values associated with these areas. For the Grant Bay, Marble River, and Spruce Bay areas, recreation facilities have been developed. However, there are currently no developments at Topknot Beach. Site developments are likely to be enhanced at Grant Bay and established at Topknot Beach once road development reaches the vicinity. The Company will also continue to develop new recreation sites in cooperation with the MoF when high valued features become accessible.

There are currently seven overnight campsites, five day-use picnic areas, and six maintained trails within the vicinity of TFL 6. Numerous other unimproved trails and recreation sites also exist. Established recreation sites are indicated on WFP visitors guides of the North Island area, which are distributed free of charge, a copy of which is found in Appendix XII. There are several Provincial Parks within and adjacent to TFL 6 (Figure 13). WFP maintained roads on both Crown and private land provide access to these areas.



Figure 13 – Provincial Parks, Reserves and Recreational Sites

Over the past five years more than \$360,000 have been spent on developing and maintaining recreational opportunities in the TFL. In 1999 there were an estimated 35,000 user days of recreation (Table 11). The majority of the funding for site development and maintenance has been born by the company. However, the BC Forest Service and Forest Renewal BC have contributed some funding towards this work. New recreation site development and maintenance will be carried out in partnership with the BC Forest Service. The level of work performed is dependent on funding available from the Company, the BC Forest Service and Forest Renewal BC. During the planning period, WFP will estimate and report, via the annual report, use days by activity category and any changes to the status of recreation facilities managed within the TFL.

Visual aesthetics are an important component in harvest planning. Known scenic areas and associated management objectives have been identified for viewsapes adjacent to waterways, travel corridors, communities and major recreation sites. Harvest blocks in scenic areas will be designed to meet the recommended visual quality objective for the area. Visually effective greenup (VEG) will be determined on a site-specific basis when new harvest blocks adjacent to previously harvested areas are proposed. For planning purposes VEG is assumed to be 5.7 meters for Block 1 and 4.1 meters for Block 2. This difference in height is due to the difference in topographic relief between the blocks.

Table 11 – Recreation Activities Summary – Estimated User Days, TFL 6 1995-1999

Location/Activity	Usage					
	1995	1996	1997	1998	1999	Total
Tourist/Recreation Site Maps	4,000	4,000	4,000	2,000	5,400	19,400
Beach Use	1,600	3,150	2,772	2,550	2,650	12,722
WFP Sites and Trails	10,700	10,350	11,600	13,900	11,100	57,650
Hunting	2,100	2,200	2,750	1,900	1,900	10,850
Fishing (Freshwater)	3,050	3,100	3,955	3,550	3,550	17,205
Fishing (Saltwater)	8,200	8,050	7,072	4,500	4,700	32,522
Firewood Cutting	3,325	3,625	3,534	1,930	2,030	14,444
Food Gathering	1,200	1,225	1,075	850	825	5,175
Kayaking	775	1,550	950	825	825	4,925
Whale Watching	1,020	1,020	1,020	1,050	1,010	5,120
Hiking and Caving	650	1,250	1,150	1,150	1,150	5,350
TOTAL	36,620	39,520	39,878	34,205	35,140	185,363

4.1.1.10 Research and Development *Element*

Objective	Indicator
<i>Support research and development</i>	<i>\$ invested in research</i>

WFP's research and development program will investigate and study new and evolving techniques and prescriptions that maintain or improve the health and productivity of forest ecosystems values while sustaining a high level of environmental protection. The company will continue to cooperate with the MoF, Canadian Forest Service, FERIC, universities and other research specialists on various research initiatives. Project funding support is often provided for by government agencies such as Forest Renewal BC.

Current research projects include:

- Continue participation in the Salal Cedar Hemlock Integrated Research Program (SCHIRP) looking at the regeneration, growth dynamics and ecology of salal dominated ecosystems.
- Continue selection breeding and testing of yellow cedar, Sitka spruce, western red cedar and western hemlock to provide healthy, well-adapted, genetically diverse superior trees for reforestation.
- Continue cooperative research on important wildlife species including Queen Charlotte goshawk (MOELP), marbled murrelet (SFU), Roosevelt elk, black tail deer, black bear, and bald eagle.
- Continue growth and yield research in line with the priorities established by the Coastal Forest Productivity Council. Particular emphasis will be placed on obtaining data for species with identified information gaps and partial harvesting systems.

- Alternate silviculture and harvesting systems will be investigated in cooperation with research agencies such as FERIC. Research topics will include productivity, cost and impacts on non-timber resource values.
- Forest productivity and health of ecosystems will be investigated. Fertilization effects on western red cedar, balsam and western hemlock stands will continue.
- Continue research into windthrow management including development of better prediction models and mitigation techniques.

As part of WFP's SFM principle an objective has been established regarding the company's support for research and development. Activities and expenditures in research and development will be reported in the annual report.

Appendix IX provides additional information on current WFP research projects.

4.1.1.11 Contribution to Provincial Revenues

Element

Objective	Indicator
Contribute to provincial revenues	Payment of fees

Timber harvest and associated forestry activities on TFL 6 contribute to provincial revenues in a number of ways. Stumpage and royalty payments for the TFL 6 harvest are significant. For the period 1995-99 over \$208.9 million has been paid. Of this total \$32.7 million was super stumpage which goes to support Forest Renewal BC and Forest Practices Code enforcement. Other fees such as AAC rentals, fees for Special Use Permits, licences of occupation, road use agreement, etc. amount to more than \$500,000 per year. All of these charges and fees go to general revenue to help pay for roads, hospitals, schools and services enjoyed by all British Columbians.

In order to fulfill their social responsibilities and meet with their SFM objectives, WFP will pay all applicable fees related to its operations in TFL 6. Payments will be reported in the annual report.

4.2 Environmentally Appropriate Forest Management

Principle

4.2.1 Conservation of Biological Diversity

Criteria

Biological diversity (or biodiversity) is the diversity of plants, animals and other living organisms in all their forms and levels of organization, and includes the diversity of genes, species and ecosystems, as well as the evolutionary and functional processes that link them (FPC, Biodiversity Guide book, September 1995). An ecosystem management approach that maintains suitable habitat conditions for native species will be used as a surrogate to maintaining biodiversity. Both coarse and fine filter approaches will be utilized to achieve biodiversity objectives.

Coarse filter strategies include the maintenance of riparian habitats, areas of old growth and a variety of patch sizes across a landscape. At the stand level coarse filter strategies include maintaining wildlife tree patches, snags, downed woody debris and deciduous trees.

Fine filter strategies deal with the identification and maintenance of habitats for known sensitive plant and animal species. Special consideration will be given to habitat requirements for red and blue listed species. Measures, which may be used to achieve fine filter strategies, include:

- 1) the creation of Wildlife Habitat Areas
- 2) establishment of Old Growth Management Areas in critical habitat zones
- 3) use of interim measures, as prescribed in the Identified Wildlife Management Strategy to protect red and blue listed species and their habitats.
- 4) Further work to be completed as part of the biological rationale for the company's FSC certification program.

4.2.1.1 Ecological Diversity

Element

Objective	Indicator
<i>Maintain a dynamic distribution of habitat over a landscape</i>	<i>Average opening size</i>
	<i>% of landscape unit reserved</i>
	<i>seral stage distribution (5 year basis)</i>
	<i>patch size distribution - % of landbase in patches > 200 ha (5 year basis)</i>
	<i>\$ invested and type of environmental inventories and assessments</i>
	<i>Timber harvesting landbase status</i>

Through planning and forest management activities, WFP will strive to maintain a variety of ecosystems across the landscape to preserve species through the provision of suitable natural habitat. In the early 1980's a comprehensive ecological classification of Tree Farm Licence 6 was undertaken by Terence Lewis (Table 4). This ecological classification provides a solid basis for interpretation of forest ecology and the response of ecosystems to management, and is a useful tool for predicting the general location of rare species, plant communities and various habitats. Using GIS tools, ecosystem representation will be analyzed and tracked at the landscape level.

In 1999 Dr. Lewis reviewed the Conservation Data Centre list of Red and Blue Listed ecosystems. Table 12 lists those ecosystems that are most likely to be located in North Vancouver Island Region. With the exception of two associations (CWHvm1 07 and CWHvm1 09), old seral stages of all ecosystem associations are well represented within existing protected and constrained areas. While Lewis has given some indication on the presence of these associations in NVIR some further inventory work may be required for the sensitive ecosystem associations so that their presence is confirmed and management efforts can be measured. Some ecosystem associations may require the implementation of specific management strategies (including possible restoration activities) to ensure that adequate amounts of these areas are well represented into the future.

Table 12 – Red and Blue Listed Ecosystems Potentially in NVIR

PLANT COMMUNITY	BEC UNIT	PROV. RANK ¹	PROV. LIST	DISTRIBUTION
VERY WET HYPERMARITIME CWH				
<i>Picea sitchensis</i> / <i>Maianthemum dilatatum</i>	CWHvh1/08	S2	RED	Limited to higher terraces on larger rivers.
<i>Alnus rubra</i> / <i>Maianthemum dilatatum</i>	CWHvh1/10	S3	BLUE	Early seral community limited to active floodplains of larger rivers; somewhat similar to WFP's S4 ecosystem.
<i>Picea sitchensis</i> / <i>Gaultheria shallon</i>	CWHvh1/14	S3	BLUE	Very limited extent. Dry, rocky, coastal headlands within 100-200 metres of the coast.
<i>Picea sitchensis</i> / <i>Kindbergia oregana</i>	CWHvh1/15	S2S3	BLUE	Very limited extent on old, raised beaches in close proximity to present-day sand/shingle beaches.
<i>Picea sitchensis</i> / <i>Malus fusca</i>	CWHvh1/19	S2S3	BLUE	Very limited extent bordering small minerotrophic (fen) wetlands (S11 ecosystems) on fluvial terraces.
VERY WET MARITIME CWH				
<i>Tsuga heterophylla</i> - <i>Pinus contorta</i> / <i>Cladina rangiferina</i>	CWHvm1/02	S2S3	BLUE	Very limited extent on very thin-soiled rock outcrops, usually in close proximity to the east coast and inner coast of Quatsino Sound, and on small offshore islands. Not operable. This includes WFP's S2P ecosystem.
<i>Tsuga heterophylla</i> - <i>Thuja plicata</i> / <i>Gaultheria shallon</i>	CWHvm1/03	S2S3	BLUE	Widespread but generally not extensive ecosystem of rocky outcrop sites with virtually continuous Folisol soils. Partly operable because of low volume, and only when closely associated with adjacent better sites/stands.
<i>Thuja plicata</i> – <i>Tsuga heterophylla</i> / <i>Polystichum munitum</i>	CWHvm1/04	S3?	BLUE	Limited to relatively well-drained colluvial, limestone and raised beach sites (inner, not outer coast), in places with some Douglas-fir. This includes WFP's mapped S3B, S12F and only a minor component (est.5%) of mapped S13
<i>Abies amabilis</i> – <i>Thuja plicata</i> / <i>Rubus spectabilis</i>	CWHvm1/07	S3	BLUE	On moist, rich sites of high fluvial terraces, fluvial fans, gullies and seepage sites; best expressed on high terraces of larger rivers (but on older surfaces and older soils than CWHvm1/08).
<i>Picea sitchensis</i> / <i>Rubus spectabilis</i>	CWHvm1/09	S2	RED	A later seral alluvial site on higher terraces of major rivers; included with WFP's S3 ecosystem on Fp and Ft terrain types.
<i>Tsuga heterophylla</i> - <i>Thuja plicata</i> / <i>Gaultheria shallon</i>	CWHvm2/03	S2S3	BLUE	Widespread but generally not extensive ecosystem of rocky outcrop sites with Folisol soils.

¹ S2 - Imperiled

Imperiled provincially because such communities with old-growth or primary forest are extremely rare or because of some factor(s) making them especially vulnerable to disturbance.

S3 – Rare or Uncommon

Provincially rare or uncommon in an old-growth / primary forest condition and susceptible to disturbance.

Within the TFL, areas in Varney, Spruce Bay, and Misty Lake have been identified as having unique ecological values. The 120 hectare Varney Gene Pool Reserve was established by the company to protect one of the most northerly sites of Douglas fir on Vancouver Island. The Spruce Bay genetic reserve encapsulates large old growth Sitka spruce within the Spruce Bay Old Growth Reserve. The unique giant black stickle- back (fish) of Misty Lake are protected within the Misty Lake Ecological Reserve and Provincial Park. The park is being removed from the TFL and the other areas are excluded from the harvesting land base.

A second component of ecosystem diversity is seral stage. Currently more than 40% of the timber harvesting landbase is covered with mature or old seral stages. Over time a trend toward a more balanced age class distribution will occur as a result of harvesting. Spatial modeling tools will be utilized to ensure the harvest pattern maintains sufficient quantities of natural habitat at each seral stage across the landscape. Areas of old growth, as defined by the FPC Biodiversity Guidebook, will be designated as Old Growth Management Areas (OGMAs) under Landscape Unit Planning. This process began in May 2000 in cooperation with the Ministry of Forests and the Ministry of Environment, Lands and Parks. It is anticipated that the process of identifying OGMAs and their specific objectives will be completed in 2002 for the Plan area.

A third component of ecosystem diversity is patch size. Wind and to a much lesser extent fire resulting from lightning strikes, are the main natural disturbance factors in the forests of TFL 6. Disturbance events have historically created a range of patch sizes up to 250 ha. As per the FPC, Biodiversity Guidebook recommendation, development will attempt to achieve a distribution of patch sizes across the landscape of 30-40% in patches <40ha, 30-40% in patches 40-80 ha and 20-40% in patches 81-250 ha.

Protected areas adjacent to and within TFL 6 are mapped. Other areas reserved from harvest, as per the FPC, include riparian reserve zones on fish streams, ungulate winter ranges and environmentally sensitive areas. With respect to riparian areas, road construction will be carefully monitored to reduce sediment where stream crossings are required. Timing windows will be observed for road construction around fish streams. Where required appropriate riparian management areas or buffers will be established to maintain the integrity of riparian reserves.

SFM indicators for area reserved, seral stage distribution, patch size distribution and environmental inventories and assessments have been established which will guide planning and forest management activities. These indicators will be reported on in the management plan and other pertinent reports.

4.2.1.2 Species Diversity

Element

Objective	Indicator
Maintain viable populations of native species	# of known species classified as threatened, endangered or vulnerable (COSEWIC, CDC) ha maintained as stand level reserves cumulative ha of wildlife habitat reserves % of area declared mixed species regeneration # trees planted by species # salmon fry released

“A Strategy for Managing Wildlife” in TFL 6 (Appendix XXII) was developed by consultant Laurie Kremsater in late 1999. The wildlife strategy for North Vancouver Island Region (NVIR) focuses on maintaining terrestrial forest dependent organisms across the tenure. To do this the Company will employ Kremsater’s three main tactics.

1. Maintaining the amount, distribution and heterogeneity of habitat and landscape elements important for biodiversity over time.
2. Ensuring representation of ecologically distinct habitat types across NVIR, to maintain lesser known species and ecological functions.
3. Maintaining the distribution and abundance of known sensitive species in NVIR.

The first two tactics will combine to form a coarse filter, which should provide for the majority of species whose needs are relatively unknown. While maintaining a range of suitable habitats across the landscape will provide for the majority of the species, the final tactic will form a fine filter to provide for those species who’s needs are relatively known and are not provided for under the coarse filter. The fine filter would provide for species on the provincial red and blue lists¹ or species identified in the Identified Wildlife Management Strategy as regionally important that warrant special consideration (Table 13). Particular attention will be given to old growth dependent species. While Kremsater and earlier work by Donald Blood (1996) have given some indications on the presence of these species in North Vancouver Island Region further inventory work is needed for many sensitive species so that a baseline can be established to measure management efforts.

¹ Ministry of Environment classification. Red-list includes species or subspecies considered to be extirpated , endangered or threatened in BC. Blue-list includes any indigenous species or subspecies considered to be vulnerable in BC.

Table 13 – Red, blue listed and regionally important Mammals and Birds

Red-listed Species	Habitat
Pacific water shrew	Riparian areas and wetlands
Keen's long-eared myotis	Caves and nearby old growth
Northern goshawk <i>laingi</i>	Low to mid elevation mature to old growth stands (nesting)
Marbled Murrelet	Old growth with large branches greater than 15cm in diameter
Blue-listed Species	Habitat
Ermine	Riparian areas with dense understory
Vancouver Island Wolverine	High elevation stand types
Roosevelt Elk	Low elevation old growth (winter) and young second growth
Great blue heron	Deciduous stand types, wetlands and estuaries
Peregrine falcon <i>pealei</i>	Rocky ledges with overhanging vegetation (nesting)
White-tailed ptarmigan	Generalist, uses a range of seral stages
Western screech owl	Low elevation riparian areas (nesting)
Northern pygmy owl	Low elevation, late seral stage stand types (nesting)
Huttin's vireo	Low elevation, late seral stage stand types (nesting)
Sandhill crane	Wetlands, estuaries
Regionally Important Species	
Black bear	Generalist, uses a range of seral stages
Black-tail deer	Generalist, uses a range of seral stages

Reforestation programs also impact on species diversity. WFP will reforest with a variety of ecologically appropriate native conifer species. Native deciduous species will also be retained within stands (minor component), riparian reserves and wildlife tree patches.

Terrestrial ecosystem restoration activities funded by FRBC are beginning to be used within the TFL. In 1999 and 2000 a pilot program was completed within the Keogh watershed of TFL 6. This pilot has lead to an expansion of similar restoration activities in the riparian areas of several other watersheds within the TFL. The pilot was designed by Vince Poulin to accelerate the recovery of forest characteristics needed for the ecological restoration of riparian habitat, water quality and channel stability. Overstocked alder and conifer stands were spaced, topped, girdled and several stems were modified (cavities cut, openings and other wounds created) to enhance the biodiversity of the area and create different types of snags for birds, bats and small mammals to use. At a number of locations trees were felled directly into the river to create debris jams and redirect water to create suitable fish spawning and rearing habitat.

Salmon enhancement programs contribute to the maintenance of salmon stocks. WFP supports three salmon enhancement hatcheries within TFL 6. They are located on the Marble River, Colonial Creek, and Cordy Creek. Over the history of salmon enhancement projects within the TFL, WFP has released over 10,880,000 fry from these facilities.

SFM principle indicators have been established to track numbers of known rare and endangered species, ha of stand level reserves, cumulative ha of wildlife habitat reserves, ha of mixed species regeneration, number of seedlings planted by species, and number of salmon fry released. Performance will be included in the annual report.

4.2.1.3 Genetic Diversity

Element

Objective	Indicator
Maintain the variation of genes within species	% seed orchard seed used

A range of suitable habitats across the landscape will contribute to maintenance of genetic diversity within plant and animal populations. Special measures may be needed in situations where research indicates that distinct local populations are threatened through habitat loss. Further study will be undertaken to quantify and describe these issues.

Reforestation and tree improvement programs contribute to the maintenance of genetic diversity for commercial conifer species. Seed orchards are designed to include a wide genetic base. Figure 14 illustrates the company's tree improvement programs. Wild seed collections are obtained from a large number of trees to ensure a wide range of genes are incorporated in the reforested area.

A SFM principle indicator for maintaining the variation of genes is the percentage of seed orchard seed used. This indicator will be reported in the annual report and other research publications.

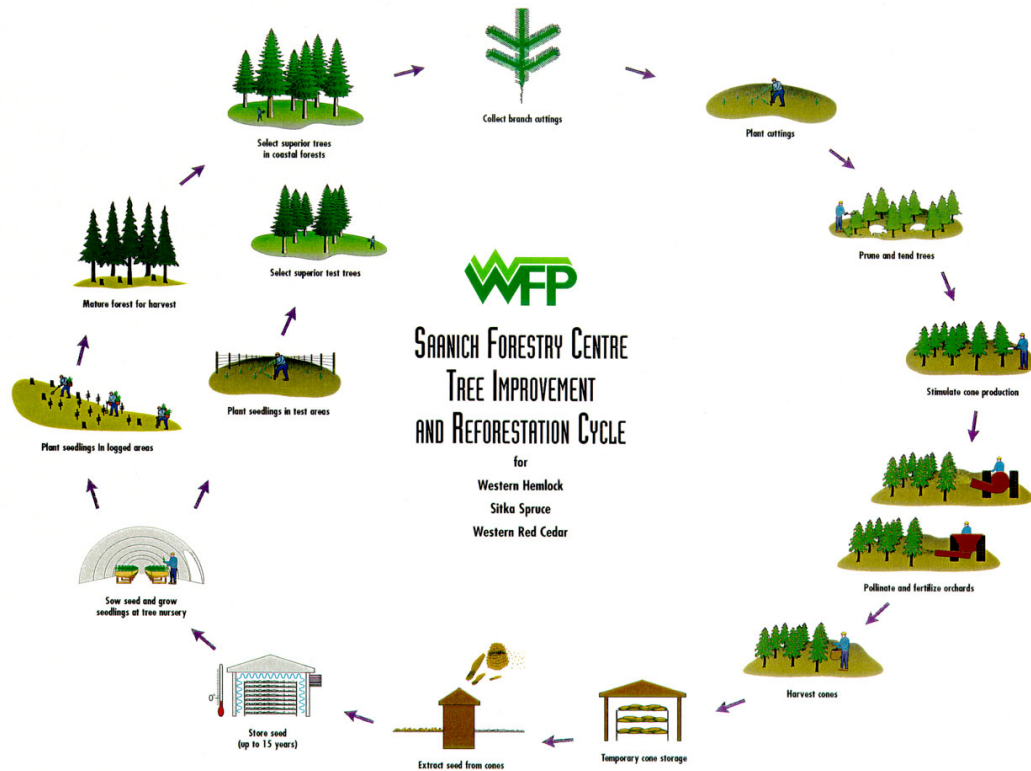


Figure 14 – Tree Improvement and Reforestation Cycle

4.2.2 Maintenance of Ecosystem Condition and Productivity

Criteria

4.2.2.1 Forest Health

Element

Objective	Indicator
Maintain forest health	pest attack status ha or # of weevil resistant spruce planting stock planted in high hazard zones # of accidental fires/ha burned

Forest health is a key element in the maintenance of ecosystem condition and productivity. WFP will maintain a forest health program that is designed to protect the licence area from damage by insects, disease, abiotic factors and fire. A proactive strategy will be pursued in detection and control. The goals of the program are to:

- Minimize the loss of timber in operable merchantable stands;
- Maintain the forest productivity and health of immature stands by monitoring insects and disease activity and implementing control action when needed;
- Salvage insect, disease, wind or fire damaged merchantable timber quickly;
- Minimize the time between felling, harvesting and processing of timber

The forests of TFL 6 have been relatively free of major insect or disease infestations. There have been no major catastrophic outbreaks causing significant unsalvaged

mortality or volume losses. The main active agents have been various defoliators and bark beetles. The last defoliator outbreak was in 1997/98 by western black-headed budworm (*Acleris gloverana*) in many stands throughout the TFL.

Spruce weevil (*Pissodes strobi*) has severely infested Sitka spruce plantations to the point where now the species is a very minor component in reforestation programs. Work is underway to produce resistant planting stock and WFP will again utilize this species on a larger scale when resistant seed or vegetative material is available.

Hemlock dwarf mistletoe is widespread throughout merchantable size mature stands. Sanitation treatments of advanced regeneration are sometimes required to prevent the spread in newly regenerated western hemlock stands. Usually vigorously growing, fully stocked stands are not impacted significantly by hemlock dwarf mistletoe.

Root diseases sometimes result in small pockets of mortality. Reforestation with less susceptible species can be used in root rot pockets

Ambrosia beetle attack degrades value of logs in inventory or storage. Keeping felled and bucked inventories low and processing logs quickly is the best method of control. Pheromone baited traps are sometimes used around dryland sort and booming grounds to trap adult beetles. Table 14 lists common insects and diseases found in TFL 6.

Table 14 – Common Insects and Disease found in TFL 6

Agent		Occurrence	Incidence	Susceptible Species ²	Management Risk
Root disease					
Armillaria root rot	<i>Armillaria ostoyae</i>	Occasional	Light	Fd, Hw, Ba, (Cw, Ss)	M
Annosus root rot	<i>Heterobasidion annosum</i>	Occasional	Light	Fd, Cw, (Hw, Ba, Ss)	L
Phellinus Root rot	<i>Phellinus weirii</i>	Infrequent	Light	Fd, Ba, (Hw, Ss) Cw	L
Stem/Branch Diseases					
Hemlock Dwarf Mistletoe	<i>Arceuthobium tsugense</i>	Frequent	Medium	Hw, (Ba, Ss)	L
Insects					
Spruce terminal weevil	<i>Pissodes strobi</i>	Common	Heavy	Ss	H
Western black-headed budworm	<i>Acleris gloverana</i>	Periodic	Light - severe	Hw, Ss, Ba	M
Ambrosia beetles	<i>Trypodendron lineatum</i>	Common	Heavy	(Hw, Ss, Ba, Fd)	H

Detection of insect and disease activity normally occurs with the collection of silviculture prescription field data and regeneration surveys. Regular aerial reconnaissance flights are also conducted to monitor health conditions. Completed silviculture prescriptions will include proposed actions for dealing with insects or

² Species are listed in order of susceptibility. Brackets indicate species are equally susceptible.

diseases noted. Any increased incidence of insect or disease activity observed during the course of regular operations will be dealt with by a specific action plan. WFP will seek assistance from specialists at the Canadian Forestry Service, MoF, universities, and consultants, if required.

If specific control measures are warranted, such as spraying to control severe defoliation outbreaks, a plan will be developed in consultation with the MoF, First Nations and other shareholders. Where control is required and biological insecticides exist, they will be favoured over non-biological treatments.

Abiotic factors such as windthrow, snow press, drought and sunscald are also considered in the forest health program. Where there is evidence of historic windthrow activity, assessments will be made during the block layout phase. Blocks will be designed with wind firm edges or treatments such as feathering or pruning will be prescribed to reduce the risk of blow down in neighboring stands. Where significant amounts of blow down occur salvage operations will be implemented to recover merchantable timber.

Species and seedlot selection combined with stocking control will be utilized to reduce damage caused by snow press, drought or sunscald.

A third component in the forest health program is fire protection. As the licence area is situated on northern Vancouver Island in the wetter maritime zone, typically the fire danger very rarely reaches extreme. WFP will comply with regulations under the Forest Act and Forest Practices Code Act relating to the prevention, detection and suppression of fires.

A Fire Preparedness Plan will be submitted by April 1 each year to the MoF. The plan will outline proposed activities, key contacts, shut down criteria, suppression equipment and company regulations with respect to fire prevention and suppression. Fuel management planning will be incorporated into Forest Development Plans and will include an analysis of anticipated fuel build-up over the five-year planning period and strategies to reduce the build-up and mitigate risk. A sample Fire Preparedness Plan is found in Appendix VII.

SFM principle indicators have been established for forest health and include: pest attack status, ha or number of weevil resistant spruce stock planted in the high hazard zone, and the number of accidental fires and ha burned

Indicators will be reported on in the annual report, Forest Development Plan and other pertinent reports.

4.2.2.2 Forest Ecosystem Resilience

Element

<i>Objective</i>	<i>Indicator</i>
<i>Maintain forest ecosystem resilience</i>	<i>NSR balance</i>
	<i>ha reforested</i>
	<i>\$ invested in basic forestry</i>
	<i>% of area surveyed free growing</i>
	<i>Average regeneration delay</i>

The functionality of forest ecosystems is paramount in our harvest planning and silviculture activities. Prior to harvesting, an ecologically based silviculture prescription (SP) specific to each cutblock is prepared by Company RPFs and submitted to the District Manager for approval. Incorporated into the SP are the results of various assessments completed by Company staff and external professionals. Assessments that may be completed include:

- terrain stability
- visual impact
- riparian
- archaeological impact
- gully
- karst
- windthrow
- forest health
- wildlife impact

Key elements of the SP in addition to the resource assessment recommendations include:

- ecosystem classification for the area covered by the prescription
- the harvest method and hazard rating for soil compaction, erosion and displacement
- the location of roads and plans for their future use
- a description of the silviculture system used and the purpose and function of trees left standing
- a description of any known non-timber resource features and plans to protect them
- the location of streams, wetlands and lakes and plans to protect them
- a reforestation plan which states the target number of healthy, well-spaced trees by species that will be achieved in the free growing stand.

The implementation of the SP is linked to WFP's basic silviculture program, the goals of which are:

- Establish a free growing stand of ecologically appropriate, well-spaced trees within the prescribed period.
- Reforest all cutblocks within 3 years after harvest to comply with prescribed regeneration delays.
- Manage regenerated cutblocks to target stocking standards

To achieve these goals the following activities will be undertaken as necessary:

- site preparation, planting, regeneration monitoring and surveys, brushing and weeding
- replanting
- basic spacing if required for declaration of free growing

WFP's basic silviculture program will conform to the FPC, Silviculture Practices Regulation. A description of the basic silviculture program is as follows:

Site preparation is not required for initial reforestation in most areas with the exception of spot accumulations of yarding slash along roads. However, in some areas containing very heavy concentrations of cedar slash, broadcast site preparation is required. This is carried out using either prescribed burning or mechanical methods. Roadside accumulations that prevent planting to desired stocking densities will be piled. Piles may be burned or left as coarse woody debris and/or wildlife habitat. Roadside accumulations account for less than 1% of the reforestation area.

Planting is the main method of reforestation and is normally supplemented by natural regeneration. Local seed is utilized for planting stock. To achieve desired stocking density trees are planted in almost every cutblock. The number of trees planted will depend on the amount of advanced regeneration retained after harvesting, expected natural seed in, brush hazard, ecosystem class and browse hazard. Seedlings are obtained from a number of container nurseries located on Vancouver Island including WFP's nursery in Saanich. Wherever available, "A" class improved seed is used. Much of the Hw, Cw and Fd seed used is "A" class and produced at WFP's localized seed orchards in Saanich. Yellow cedar stockings are produced from rooted cuttings. Wild seed collections or purchases are made to address any deficiencies in short-term seed supplies. A ten year supply of seed is normally maintained in storage for most species.

Mechanical site preparation and/or spot applications of herbicide may be required prior to fill-planting. Herbicide applications are planned as per Pest Management Plans or Pesticide Use Permits, which are discussed under brush, control. Historically less than 4% of the reforestation area requires fill-planting.

Silviculture surveys are performed on all regenerating cutblocks to monitor progress towards achieving free growing status. Information collected includes species; total number of stems and number that are well-spaced; brush species and distribution; and qualitative remarks. The results are used to prescribe follow-up silviculture treatments if necessary. Generally one survival survey and a minimum of one regeneration survey are undertaken prior to a final free to grow assessment. Survey results are reported to the MoF via Major Licence Silviculture Information System (MLSIS) reports.

An integrated pest management approach will be used when determining treatment thresholds, and treatments for brush control. The majority of TFL 6 is covered under either a five year Pest Management Plan (Jeune Landing Operation) or Pesticide Use Permits (Holberg Operation) which are approved by the MOELP. These documents specify when and how brush control treatments will be undertaken, consultation protocols, environmental protection measures and monitoring. The Company is striving to reduce the amount of pesticides used. Where feasible, alternatives to herbicides are used to control brush. Table 15 indicates the Five-Year Basic Silviculture Goals (hectares) for TFL 6.

Silviculture Standards for TFL 6 are in Appendix XIII.

Table 15 – Five-year basic silviculture activity goals (ha) for TFL 6

Activity	Previous 5 Year Avg.	2001	2002	2003	2004	2005
SP's (Area Logged)	1450	1500	1500	1400	1400	1400
Site Preparation	30	30	25	25	25	25
Planting	1350	1300	1200	1200	1200	1200
Silvicultural Surveys	2120	3000	3000	2500	2500	2500
Brushing	1170	1000	850	800	700	700
Free Growing Surveys	3300	2000	1500	2000	1500	1500

There has been discussion on the requirements to retain coarse woody debris (CWD) throughout the management area. CWD retention levels could possibly be used as an indicator of forest ecosystem resilience. Over the life of this plan WFP will work with government agencies to develop a strategy for the retention of CWD. Where necessary WFP will also evaluate the necessity to develop a recruitment plan for CWD.

Performance on the indicators for the maintenance of forest ecosystem resilience will be reported in the annual report.

4.2.2.3 Forest Ecosystem Productivity

Element

Objective	Indicator
Maintain forest ecosystem productivity	ha harvested by silviculture system # of growth and yield plots established and maintained km of fish habitat created/enhanced ha of riparian restoration \$ invested in enhanced forestry ha fertilized, spaced and pruned

Past harvesting in TFL 6 is providing a variety of patch sizes and shapes across the landscape. With the advent of the Forest Practices Code of BC Act (FPC) more attention is being given to riparian protection, coarse woody debris and maintenance of structural forest attributes important for biodiversity function.

Harvesting during the planning period will continue to be mainly in old growth forest stands more than 120 years old. Some 80-120 year old second growth of wind origin is also being harvested where it fits in with the development of older stands or where specific objectives at the landscape level dictate. For example the creation of spring forage around ungulate winter ranges may result in second growth harvest.

A variety of silvicultural systems are being utilized all of which are variations of clearcutting or partial cutting as defined by the FPC, Operational Planning Regulation (OPR). The primary silviculture systems used to date in the TFL have been patch clearcutting and patch clearcutting with reserves. Other silviculture systems, which have been utilized, include single tree and group selection. Recently the retention silviculture system, which maintains structural diversity over the cut block for a

minimum of one rotation is being utilized. WFP is currently investigating alternate harvesting systems and has begun a Retention Silviculture System Trial to analyze the practicability of using this system within the Plan Area. WFP remains committed to using silviculture systems that maintain forest ecosystem productivity and function.

As discussed under the section Research and Development, WFP maintains permanent growth and yield plots which are used to measure stand productivity. New plots will be established to monitor incremental silviculture treatments in line with priorities of the Coastal Forest Productivity Council.

WFP has been very active in the creation and enhancement of fish habitat through the placement of in-stream structures and construction of off-channel habitat. These programs have been funded through the FRBC Watershed Restoration Program. Within TFL 6 over 20 km of streams have been enhanced over since 1996 and several off-channel habitat projects have been undertaken. These programs contribute to the restoration, maintenance and enhancement of local fish populations.

As described in Section 4.2.1.2 terrestrial ecosystem restoration activities are beginning to be used within the TFL. These activities were designed to accelerate the recovery of forest characteristics needed for the ecological restoration of riparian habitat, water quality and channel stability and enhance the biodiversity of the area. The expansion of this program to other watersheds within the TFL will take place with available funding during the life of this plan.

Once the commitments and obligations of the SPs have been fulfilled, stands are available for incremental silviculture treatments. WFP will strive to optimize the timber producing capacity of forest ecosystems while conserving other resource values. WFP currently has a multi-year agreement with Forest Renewal BC to fund incremental silviculture activities such as backlog brushing, backlog planting, juvenile spacing, pruning and fertilization. A Resource Management Plan has been developed and approved by FRBC and the MoF which sets out goals related to volume and value gains to be made through various incremental silviculture treatments as well as social outputs such as training and employment. Stand Management Prescriptions (SMPs) are prepared and approved by Company RPFs, which give specific treatment objectives, time frames, and address non-timber resources. SMPs are reviewed and approved by the MoF. Table 16 outlines the planned five-year goals enhanced silviculture goals based on continued FRBC funding at current levels.

Table 16 – Five-Year Enhanced Silviculture goals (ha)

Activity	Previous 5 Year Average	2001	2002	2003	2004	2005
Juvenile Spacing	460	300	300	300	300	300
Pruning	475	450	450	350	350	350
Fertilization	1769	800	1200	800	1200	800

Six SFM indicators have been established to report on Forest Ecosystem Productivity. WFP will report annually on the hectares harvested by silviculture system, number of growth and yield plots monitored, the kilometers of fish habitat created/enhanced, amount of riparian restoration, investments made on enhanced forestry, and amount of ha fertilized, spaced and pruned each year.

4.2.3 Conservation of Soil and Water Resources

Criteria

4.2.3.1 Productive Area

Element

Objective	Indicator
Minimize permanent loss of productive area	% permanent access within cutblocks

The total area of TFL 6 is approximately 198,000 ha. Of this total, 185,500 ha are considered productive. Existing roads, trails and landings occupy roughly 2,500 ha or 1% of the productive land. In 1999 a reclassification of the TFLs landbase was completed that reflects the physical accessibility and an economic availability of the land. Lands classified as physically inaccessible would include lands where there is a physical barrier or terrain constraint leaving accessibility virtually impossible mostly due to environmental sensitivity. Approximately 11,600 ha of productive area was identified and were removed from the timber harvesting landbase due to physical accessibility. Approximately 950 hectares have been removed from the timber harvesting landbase as economically unavailable under current conditions.

One of the primary goals of harvest planning is to minimize impacts on the net productive landbase. Permanent and temporary access structures are planned to minimize the total amount of road required to safely and economically harvest the timber. For the period 1995-99 the percent of permanent access within conventional cutblocks has averaged 4.6%.

Mainline and branch road prisms during this period have been on average 11 metres wide. Spur roads have averaged 9 meters. Future roads are forecast to occupy 2,800 ha. The following practices are undertaken to minimize the loss of productive area.

- Sensitive road cut slopes and fill slopes are/or planted with trees and grass seeded
- Temporary access structures and abandoned roads are debuilt wherever possible
- Excess material generated by road building on slopes greater than 60% is generally end hauled or keyed in to minimize side-casting
- Historically over steepened fill slopes are rehabilitated through side cast pullback and planted with trees and grass seeded

Terrain stability assessments are completed where harvesting is proposed in areas of moderate or high hazard for landslide initiation following timber harvesting or road construction. If harvesting proceeds in these areas the recommendations made by terrain specialists are followed in plan development and implementation. Rainfall shutdown standard operating procedures are followed to ensure the safety of personnel and reduce the risk of erosion and landslides.

One SFM principle indicator has been established to monitor impacts to the productive forest landbase. It is the % permanent access within cutblocks. Results will be in the annual report.

4.2.3.2 Water Quality

Element

Objective	Indicator
Maintain water quality	km of road deactivated
	ha in Riparian Management Areas (RRZ & RMZ)
	Ha of CWAPs completed by year
	# of reportable spills to water

The watersheds in TFL 6 drain large catchment areas on northern Vancouver Island. Many support populations of anadromous fish and some supply domestic drinking water. There are currently thirty-two identified drinking water locations in TFL 6.

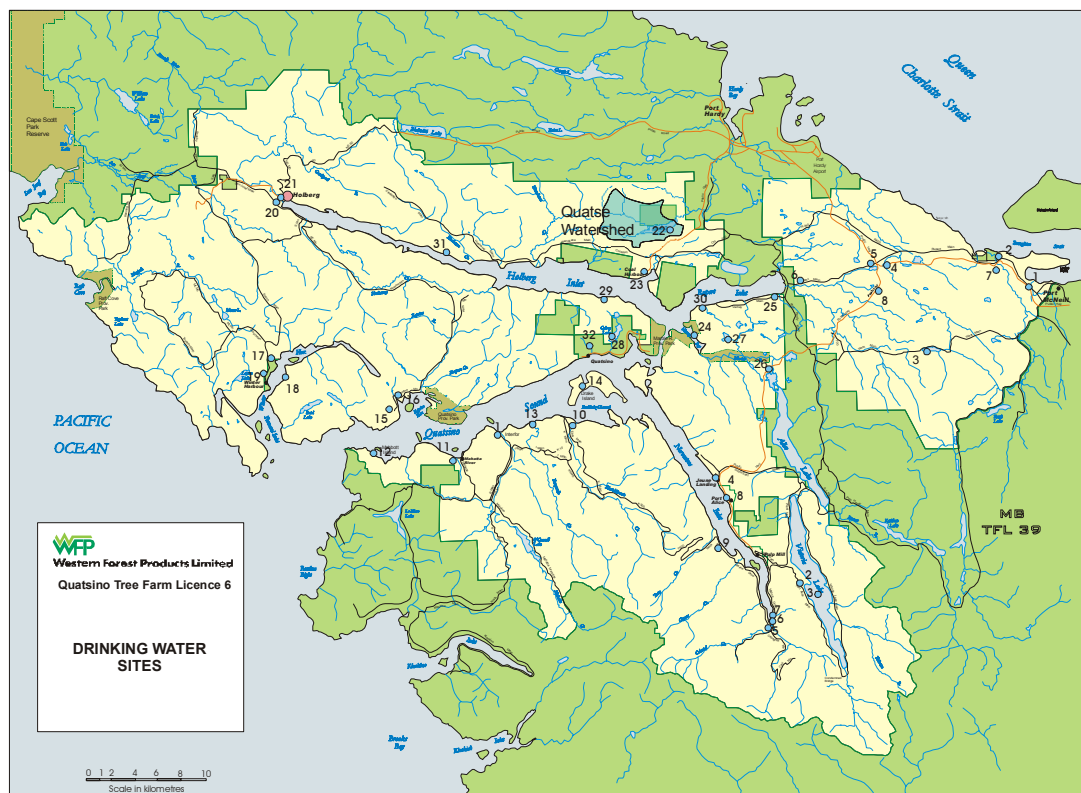


Figure 15 – Drinking Water Sources

Within the TFL there is one designated, but unused, community watershed and one non-designated watershed from which water is drawn for community use. Calbick Creek Community Watershed is located adjacent to the Quatsino First Nations Reserve at Coal Harbour. This watershed although designated under the FPC is not used. The community of Coal Harbour and the Quatsino First Nations draw their water supply from Quatsie Lake located north of the community. Although this watershed is not designated under the FPC, the Company manages it as if it were a designated watershed.

Water quantity stations are established on Simpson Creek and San Josef River. These stations are being maintained by Environment Canada – Water Survey of Canada. As well as contributing data to a larger study of streams on Vancouver

Island the stations will provide background data which will serve as a baseline to measure the impact of future harvesting in the watersheds.

FPC riparian reserve zone and management zone provisions will be incorporated into operational plans to protect water quality. In addition, WFP has standard operating procedures (SOP) designed to protect water quality and the environment in general. These standards include formal procedures for activities and maintenance of machinery in riparian management zones, application of herbicides for brush control and block design and layout. Appendix VIII provides a sample SOP.

Road deactivation, road maintenance and watershed restoration projects funded by FRBC are also key activities in the maintenance of water quality. The focus of road deactivation and maintenance is to maintain natural surface and ground water flow patterns and to minimize risks posed by road related landslides or other mass wastage events. Specific watershed restoration projects are directed at maintenance and enhancement of fish habitat. Appendix XIV provides project details and a summary of the work completed in the Keogh River Watershed.

SFM principle indicators have been established for water quality. They are: ha in RMAs (RRZs and RMZs), km of deactivated road, ha of CWAPs completed, and number of reportable petroleum spills to water. Results will be reported in the annual report and the Environmental Management System (for spills to water).

4.2.4 Contribution to Global Ecological Cycles

Criteria

4.2.4.1 Carbon Sequestration

Element

Objective	Indicator
Maintain carbon balance relative to company operations	Fuel consumption/m ³ Ha of NSR (current + backlog)

Carbon Sequestration refers to the provision of long-term storage of carbon in the terrestrial biosphere, underground, or the oceans so that the buildup of carbon dioxide (the principal greenhouse gas) concentration in the atmosphere is reduced or slowed. In some cases, this is accomplished by maintaining or enhancing natural processes.

Global carbon cycles are not fully understood. However, it is believed that the global carbon cycle is out of balance. The increased atmospheric CO₂ level is making global climatic change more likely. There has also been a general belief that the burning of fossil fuels contributes to increased atmospheric CO₂ levels. While the burning of fossil fuel is necessary to carrying out forest management activities the levels of consumption will be monitored relative to the production levels.

Establishment and maintenance of forests is an important aspect of the terrestrial carbon sequestral. As forests grow they convert carbon dioxide in the air to carbon stored in trees. Forests form a large terrestrial sink for carbon. Harvesting will be followed by prompt reforestation.

The indicators for carbon sequestration will be monitored on an annual basis and reported in the annual reports.

4.2.5 High Conservation Value Forests

Criteria

The term "High Conservation Value Forest" or HCVF originates from Principle 9 of the Forest Stewardship Council's ten Principles and Criteria. Management of HCVF focuses on the precautionary maintenance or enhancement of attributes such as significant concentrations of biodiversity, ecosystems at risk, watersheds, unstable terrain, or fundamental economic or cultural needs of local communities or First Nations.

Management of attributes identified as sensitive to forest management has been ongoing for many years. Management strategies for them have been described in previous plans or other sections herein.

TFL 6 is part of a defined forest area for the purposes of certification under the Forest Stewardship Council. A summary paper entitled "Statement on High Conservation Value Forests of North Vancouver Island Region" was drafted in 1999 and submitted to the public, stakeholders, and First Nations for review and input. From the feedback received a more comprehensive document entitled "High Conservation Forests North Vancouver Island Region" was developed and circulated for further input. The most recent version of the strategy can be found in Appendix XX. Within this strategy, proposed HCVF attributes are identified and strategies specifically developed for the management and monitoring of them. The Licensee has enlisted the services of a team of specialists (biologists & ecologists) to review this strategy and refine the approach taken on High Conservation Value Forests. It is expected that this review will result in an updating of the related appendices. This review and updating is expected to be completed by the end of 2001. This strategy is subject to continual review and updating as new information is learned, consultation continues, and society's values change over time.

4.3 Socially Beneficial Forest Management

Principle

4.3.1 Responsibility for Sustainable Development

Criteria

4.3.1.1 First Nations

Element

Objective	Indicator
<i>Provide opportunities for consultation on forest management</i>	<i>records of meetings and correspondence</i>
<i>Increase First Nations involvement in forest management</i>	<i>% silviculture contracts to First Nations other person-days employment to First Nations and/or joint ventures</i>
<i>Protect First Nations cultural features</i>	<i># of AIAs</i>

The Quatsino, Kwakiutl, and Tlatlasikwala First Nations traditional territories cover much of the TFL 6 land base as shown in Figure 16.

The Quatsino First Nation traditional territory covers about 85% of TFL 6 and includes all waters that flow into Quatsino Sound. The communities of Port Alice, Holberg,

Coal Harbour, Quatsino and Winter Harbour all lie within the boundaries of the Territory. Coal Harbour Reserve, a community of approximately 342 residents located outside Port Hardy, is the main community of the Quatsino First Nations.

The Kwakiutl First Nations traditional territory runs from Port McNeill to Port Hardy and contains all lands draining into Queen Charlotte Strait. The communities of Port Hardy and Port McNeill lie within the boundaries of the territory. Fort Rupert, the main community of the Kwakiutl (located near the Port Hardy Airport) has a population of roughly 326. An additional 265 band members live off of the reserve.

The Tlatlasikwala First Nations traditional territory covers the northern tip of Vancouver Island including several adjacent islands and all lands draining into the San Josef River. The band is currently in the process of relocating to Bull Harbour north of Port Hardy. Currently the 41 band members residences are split between Port Hardy and Whe-la-la-u Reserve in Alert Bay.

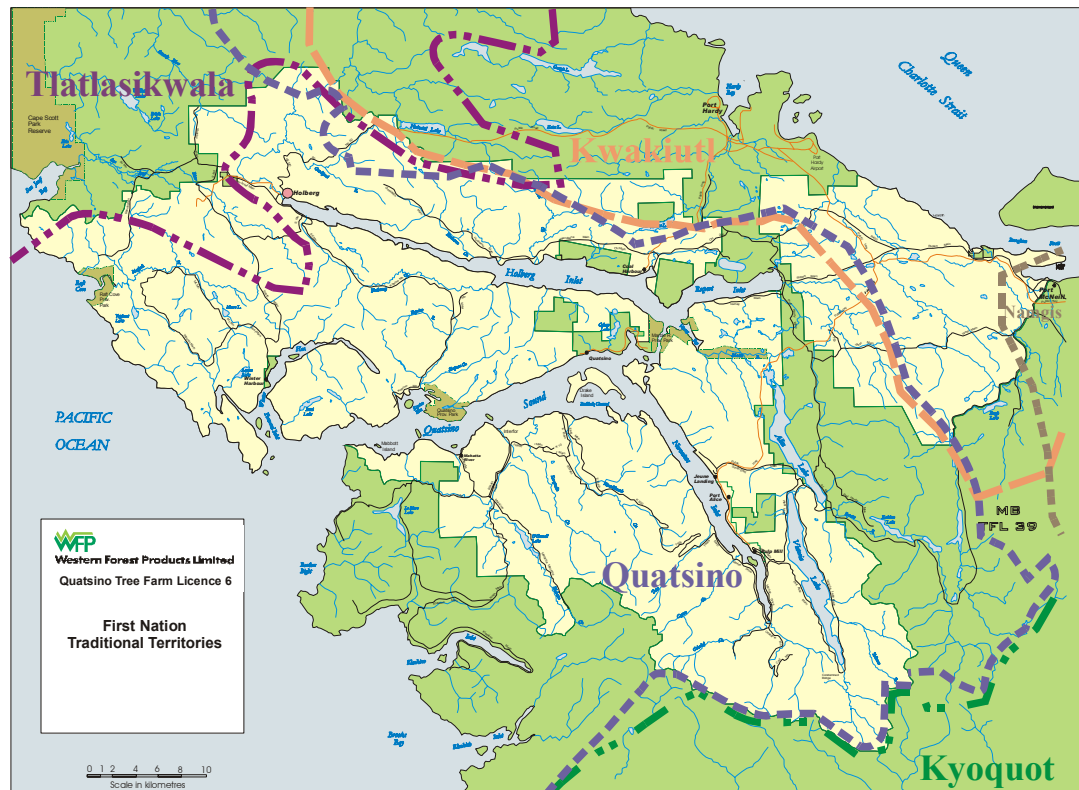


Figure 16 – First Nations Territories

WFP is committed to maintaining a meaningful and respectful relationship with First Nations. The TFL Management Plan and Forest Development Plans (FDP) are referred to the First Nations leadership for review and comment. WFP planners meet with Band Councils to review the draft plans and address concerns brought forward by the Bands. Development plans are adjusted to respect First Nations cultural and heritage resources.

An archaeological overview assessment of the TFL was completed as part of the Port McNeill Forest District overview assessment. This assessment was quite coarse in scale, however it identified some potentially significant areas. Areas with high potential, areas that are adjacent to those areas with high potential, and areas

identified in the field as having trees that have potentially been culturally modified, are surveyed by an archaeological consultant and where possible a member of the First Nations involved, to produce an archaeological impact assessment (AIA). The AIA is sent to the First Nations band involved for review and comment.

WFP has implemented a policy of First Nations consultation and involvement in harvesting and silviculture activities. WFP has encouraged the Quatsino Band's development of logging and silviculture capability. The Band holds a 30,000 m³ renewable logging contract and receives direct awarded silviculture contracts. WFP has agreed to support a Forestry Coordinator for the Quatsino Band and currently supports one for the Kwakiutl Band. WFP also encouraged the development on a Kwakiutl silviculture crew. WFP has also provides training in logging for Kwakiutl Band members, and a few of WFP's permanent employees have come out of this training initiative. WFP has established an agreement with FRBC that allows projects to be directly awarded to local First Nations contractors wherever possible, provided the bids are within the range of historic, open-tendered projects of a similar nature in the North Vancouver Island Region. Target groups for enhanced silviculture projects are First Nation companies operating within their traditional territories. The goal is to provide full employment for those crews in a variety of silviculture activities in addition to providing training opportunities for new people. Over the last five years, First Nations employment was approximately 2,050 person days per year for silviculture contracting in TFL 6.

Western Forest Products provides timber to First Nations groups for a range of traditional uses, including canoes, drum logs, totem poles, and big houses.

First Nations indicators will be monitored through a combination of annual reports and Forest Development Plans.

4.3.1.2 Communities

Element

Objective	Indicator
<i>Conduct effective consultation with communities</i>	<i># and types of contacts/meetings/consultation</i>
<i>Maintain and enhance community stability</i>	<i>\$ invested in public projects</i>
	<i>\$ spent on forest education program</i>

WFP is one of the founding members and is an equal partner in the North Island Discovery Center. The center provides forestry and tourist information to all people who visit. In conjunction with the center WFP offers the public free forestry tours of forestry and logging operations within TFL 6. Many special tours are also conducted on request. Approximately 300 people participate in the tours each year.

For several years, the company has also been providing a forestry program for several schools in the Victoria Area at our Lost Lake Seed Orchard and now our Saanich Forestry Center. Forestry and logging tours are given to several local school classes. Programs and presentations are given on an annual basis at several schools within the local communities. The school programs have been very enthusiastically received.

Visitor guide maps of the North Vancouver Island Region are popular. Approximately 5,400 copies were distributed in 1999. Appendix XII is a copy of the Visitor Guide.

The Company is dedicated to supporting local businesses and contractors. Businesses in the North Vancouver Island Region rely heavily on trade generated by local forestry operations.

The communities of the North Vancouver Island Region benefit from Western Forest Products support of many events and recreational activities. Financial support and in-kind contributions are provided for: Local Chamber of Commerces, Port McNeill Museum, Seven Hills and Port Alice Golf Courses, Tri-Port Raceway, Seven Mile Motorcross Track, Holberg Volunteer Fire Department, Port Alice Volunteer Fire Department, Port McNeill Volunteer Fire Department, Holberg First Responders, Mount Waddington Regional Fall Fair, National Forest Week, several Logger's Sports Days, several local sports teams, and several other local events and groups. WFP also has a long history in the development of community infrastructure (arena, track, pool, etc.)

Western Forest Products' employees are very active in their communities. In the last five years, employees have continued to be members of many local groups including:

- Local Government and Committees
- Mount Waddington Community Resource Board
- Special Management Zone Working Group
- Local Harbour Commissions
- North Vancouver Island Salmonid Enhancement Association
- North Island Tourism Initiative
- Canadian Institute of Forestry (Pacific Section)
- Employee Family Assistance Program
- Community Action Network
- Mount Waddington Community Health Council
- Volunteer Fire Departments
- British Columbia Ambulance Service, and the Provincial Emergency Program.

Figure 17 shows the location of operations and communities. Appendix XVII provides a profile of the main communities located within TFL 6.

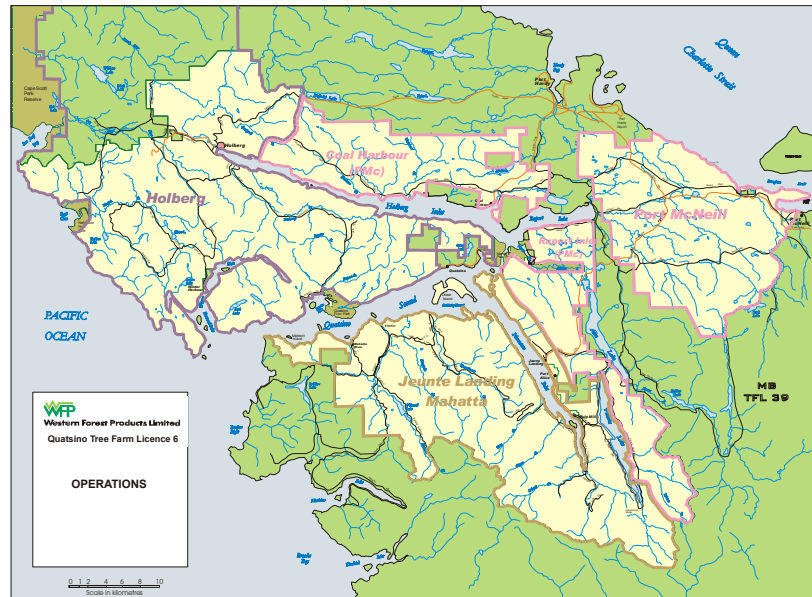


Figure 17 – Forest Operations

Indicators related to communities will be reported annually within TFL annual reports.

4.3.1.3 Employment

Element

Objective	Indicator
Sustain employment levels	Total Person-days and \$/m ³

Tree Farm Licence 6 is a significant generator of employment in logging, silviculture and forest management. Other employment is created as the logs produced in TFL 6 are processed in the Port Alice Pulp Mill and other Doman mills. Table 17 summarizes the average annual direct employment over the past five years as a consequence of operations in TFL 6.

Table 17 – Annual Direct Employment 1995 – 2000

Category	Average Annual Person Days of Employment
Planning/Harvesting	81,090
Silviculture & Integrated Use	13,630
Administration	4,250
Subtotal	98,970
Manufacturing	102,725
Total	201,695

Over the planning period WFP intends to maintain employment levels through the following strategies:

- Harvest 100% of the AAC in each cut control period
- Meet or exceed contractor requirement as per Timber Harvesting Contract and Subcontract Regulation
- Where possible expand the silviculture and watershed restoration programs through long term funding commitments from FRBC
- Maintain and where possible expand minor forest products harvesting

program

- Work with First Nations to expand employment opportunities

Employment strategies hinge on the degree of decline of the AAC, no diminishment with company tenure rights and continued or expanded funding from Forest Renewal BC over the next 5 years.

Employment levels will be reported on an annual basis within TFL annual reports.

4.3.1.4 Compliance

Element

Objective	Indicator
Provide adequate training	# of training hours
Achieve compliance with all legislation and regulations	% compliance

Forestry activities in TFL 6 are focused on sustainable development and environmental protection. Legislation, regulations, policy and guidebooks provide standards for forest practices. WFP's objective is to achieve full compliance with all legislation and regulations.

WFP has developed a series of Standard Operating Procedures (SOPs). These SOPs guide each of our staff, employees and contractors. These SOPs and associated training programs assist WFP's operations to achieve a consistent and high level of responsible forest stewardship.

The SOPs are a key element of WFP's environmental management system. The SOPs set minimum standards for specific forest management activities. Progress and performance will be assessed through internal and external audits of each operation.

Percent compliance with rules and regulations will be monitored. Contraventions will be investigated and, if required, controls and policy changes put in place to prevent reoccurrence.

Compliance indicators will be reported on an annual basis within TFL annual reports.

4.3.1.5 Employee Relations

Element

Objective	Indicator
Provide adequate training	# of training hours
Provide regular communication to employees and contractors	WFP Western Spirit and Environment Matters Newsletter circulation

Maintaining good relations with WFP's employees, contractors and their employees is an important element of WFP's Sustainable Forest Management Program. The objective is to provide adequate training and regular communication to workers and contractors. The employees are trained in sustainable forestry practices and through regular communication the employees are informed of new practices. Existing

practices are reviewed and updated as required. Training hours will be reported annually within TFL annual reports.

WFP's Western Spirit and Environment Matters Newsletters (samples in Appendix XIX), quarterly publications are circulated to employees. Training records and training needs for all employees are maintained at WFP's Regional Office. The Environmental Committees in each operation meet quarterly to review their environmental management program.

4.3.1.6 Safety

Element

<i>Objective</i>	<i>Indicator</i>
<i>Provide a safe working environment</i>	<i># of lost time accidents</i>

Western Forest Products recognizes the necessity of establishing and maintaining a safe working environment for all employees, contractors and their employees. Safety has always been and always will be an important aspect of all operations. Each operation's safety program includes the following:

- Monthly meeting of safety committees
- Develop safe work practices for each activity
- Provide training in safe work procedures
- Communicate safety to all workers on a regular basis
- Reporting of dangerous conditions and incidents

Safety performance of the company is monitored on a monthly and annual basis.

5.0 Sustainable Forest Management Plan Consultation

The preparation of Management Plan 9 for TFL 6 has followed the old TFL Management Plan process. The process laid out a 30 month time line leading up to the approval of a new plan and included the following: a public review of the current plan, submission of a review strategy for the new plan, a Statement of Management, Options, Objectives and Procedures (SMOOP)(Appendix XXIII), an Employment and Economic Opportunity report (EEO)(Appendix XXIV), an information package, a timber supply analysis and 20 year projection, and a draft MP. This process was initiated for TFL 6 in July 1998 following direction from the MoF Vancouver Region calling for a public review of the current plan.

WFP developed a public and stakeholder review strategy for MP 9 late in 1998 and carried out the public review of the current plan in August and September of 1998. Advertisements were placed in local newspapers and on the Company website inviting comments on current performance under MP 8 and suggestions for consideration. In addition, letters were sent to 250 stakeholders advising them of the review and an invitation to comment.

MP 8 was on review in Port McNeill (WFP and MoF), Port Alice, Holberg, Nanaimo (MoF), Victoria (MoF) and Vancouver (WFP, Head Office) from July 26, 1998 to September 30, 1998. Seven responses were received on the Statement of Management Objectives, Options and Procedures (SMOOP). Generally responses were very favourable and highlighted the company's history and strong community ties. One local resident was concerned with road deactivation limiting access into areas for honey collection. The resident was satisfied that several opportunities would still exist for honey collection. Other comments were related to optimizing the value of the wood produced, forest management activities above Port Alice and windthrow management. There were no significant changes required to the SMOOP.

A revised MP 9 stakeholder and public review strategy was submitted with the Draft MP as part of Appendix X. Open houses were held in Port McNeill, Port Alice, Holberg and Port Hardy. WFP staff also met with municipal councils in the above-mentioned communities and with the three First Nations whose traditional territory encompasses TFL 6. A summary report was prepared and is appended to the proposed MP in Appendix X. A total of 163 persons attended the 13 sessions. A total of 14 written comments were received as well as the Regional Manager's comments.

6.0 SUMMARY IMPACT ANALYSIS

Economic, environmental and social impacts will be monitored using indicators developed to measure WFP forest management activities. These indicators will be reported on an annual basis in the TFL Annual Report. If significant negative impacts become apparent, where appropriate, WFP will develop specific management strategies to mitigate the effects of the company's forest management activities.

6.1 Economic

One of WFP's primary principles is to be an economically viable forest products company which can compete in world markets. This will be achieved by:

- economically harvesting the licensee portion of the proposed AAC
- achieving production and administration efficiencies
- through optimal log allocations and utilization, WFP intends to improve product values and enhance lumber recoveries

Independent certification of forest operations will provide assurances that operations are being conducted in a sustainable manner.

Table 18 summarizes the Criteria, Elements, Objectives and Indicators that will be used to measure WFP's performance in practising Economically Viable Forest Management.

Table 18 – Indicators of Economically Viable Forest Management

Criteria – Multiple Benefits		
Element	Objective	Indicator
Timber Supply	Harvest AAC	% achievement of AAC, 5 year cut control
		estimated total value of timber produced
	Prevent timber loss	ha of non-recoverable losses due to wind, fire
	Efficient utilization	m ³ of billable waste
Harvest Methods	Employ appropriate harvest methods	harvest system by volume
Contractor Commitment	maintain contractor commitments	% of required contractor commitment achieved
Profitability	achieve a reasonable return on investment	margin/m ³
Non-Timber Forest Products	encourage NTFP utilization	volume by type of NTFP (m ³ , kg)
Special Forest Products	encourage utilization of special forest products	estimated total value of special forest products
Access Management	provide for public access	km of road maintained
Recreation	maintain recreation sites and Features	# and type of recreation sites maintained
		user days by category
Research and Development	support research and development	\$ invested in research
Contribution to Provincial Revenues	contribute to provincial revenues	payment of fees

6.2 Environmental

Environmentally appropriate forest management is another key principle in WFP's Sustainable Forest Management Policy. WFP is committed to the protection of the environment and the sustainable development of the resources under our stewardship. This will be achieved by implementing several forestry and environmental management practices that meet or exceed government standards. Management practices will continue to be evaluated and adopted in response to new information and experience.

It is anticipated that implementation of new environmentally appropriate management practices may have a negative impact on the AAC. However, this impact may be overcome by implementing intensive forest management in those areas where it is ecologically appropriate.

Table 19 summarizes the Criteria, Elements, Objectives and Indicators that will be used to measure WFP's performance in practising Environmentally Appropriate Forest Management.

Table 19 – Indicators of Environmentally Appropriate Forest Management

Criteria – Conservation of Biological Diversity		
Element	Objective	Indicator
Ecological Diversity	maintain a dynamic distribution of a habitat over a landscape	average opening size (ha)
		% of defined area reserved
		seral stage distribution (5 year basis)
		patch size distribution - % of landbase in patches > 200 ha (5 year basis)
		\$ invested and type of environmental inventories and assessments
		THLB Status
Species Diversity	maintain viable populations of native species	# of known species classified as threatened, endangered or vulnerable (COSEWIC & CDC)
		ha maintained as stand level reserves
		cumulative ha of wildlife habitat reserves
		% of area declared as mixed species regeneration
		# planted by species
		# fry released by species
Genetic Diversity	maintain the variation of genes within species	% seed orchard seed used
Criteria – Maintenance of Ecosystem Condition and Productivity		
Element	Objective	Indicator
Forest Health	maintain forest health	pest attack status
		ha or # of weevil resistant spruce planting stock in high hazard zones
		# of accidental fires/ha burned
Forest Ecosystem Resilience	maintain forest ecosystem Resilience	NSR balance
		% of surveyed area free growing
		average regeneration delay
		ha reforested
		\$ invested in basic forestry

Forest Ecosystem Productivity	maintain forest ecosystem productivity	# of growth and yield plots monitored
		km of fish habitat created/enhanced
		\$ invested in enhanced forestry
		ha of riparian restoration
		ha fertilized, spaced and pruned
		ha harvested by silviculture system
Criteria – Conservation of Soil and Water Resources		
Element	Objective	Indicator
Productive Area	minimize permanent loss of productive area	% permanent access within cutblocks
Water Quality	maintain water quality	km of road deactivated
		ha in Riparian Management Areas (RRZ & RMZ)
		Ha of CWAPs completed by year
		# of reportable spills to water
Criteria – Contribution to Global Ecological Cycles High Conservation Value Forests		
Element	Objective	Indicator
Carbon Sequestration	maintain carbon balance relative to company operations	ha of NSR (current + backlog)
		fuel consumption/m ³

6.3 Social

WFP is committed to respecting, understanding and supporting First Nations, community and employee aspirations for stability and certainty. An economically viable company translates into stable employment for employees and promotes community stability. Continuation of environmentally appropriate management practices will ensure long-term sustainable socially beneficial operations.

Table 20 summarizes the Criteria, Elements, Objectives and Indicators that will be used to measure WFP's performance in practising Socially Beneficial Forest Management.

Table 20 – Indicators of Socially Beneficial Forest Management

Criteria – Responsibility for Sustainable Development		
Element	Objective	Indicator
First Nations	provide meaningful consultation on forest management	records of meetings and correspondence
	increase First Nation involvement in forest management	% silviculture contracts to First Nations other person-days employment to First Nations and/or joint venture
	protect First Nations cultural features	# of AIAs
Communities	conduct effective consultation with communities	# and types of contacts/meetings/consultations
	maintain and enhance community stability	\$ invested in project projects
		\$ spent of forest education programs
Employment	Sustain employment levels	total person days and jobs/ m ³
Compliance	provide adequate training	# of training hours
	achieve compliance with all legislation and regulations	% compliance
Employee Relations	provide adequate training	# of training hours
	provide regular communication to employees and contractors	<i>WFP Western Spirit and Environmental Matters Newsletter</i> circulation
Safety	provide a safe working environment	# of lost time accidents

6.4 Impacts Due to Implementation of Management Plan 9

6.4.1 Harvest and Employment Levels

The Timber Supply Analysis (Appendix III) for MP 9 proposes an Allowable Annual Cut of 1,459,000 m³. This proposed harvest level is 2% less than the MP 8, which was 1,489,000 m³. This decrease is in line with the proposed step down that was outlined within MP 8.

Due to the increased complexity of forest management activities and the small decrease in AAC, there are no significant changes anticipated to the employment levels as a result of implementation of this plan.

6.4.2 Non-Timber Values

Management Plan 9 is the first plan to fully incorporate the results of implementation of the Forest Practices Code. MP 9 is also the first plan that has been designed to be in draft compliance with both FSC and CSA certification systems. While non-timber values were mentioned in previous MPs, MP 9 has taken a more comprehensive approach. These values include wildlife, streams, recreation, visual quality, biological diversity, non-timber forest products, soil, cultural, social, and high conservation value forests. MP 9 has also outlined in detail how each of these values will be monitored.

6.4.3 General Impacts of Implementation

WFP believes that the implementation of this plan will aid in giving the company greater economic stability. With greater economic stability, more economic opportunities would be available for the communities that are directly tied to the operations within TFL 6.

It is not anticipated that the implementation of this plan will have any further negative impacts over the current plan.